

Mitigating School Closures: Parental Responses and Tutoring Vouchers*

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Abstract

We investigate how school closures affect educational outcomes through parental responses. Exploiting quasi-experimental variation in school-level closure days and a linked administrative–survey panel, we find that ten additional school closure days increase private tutoring expenditure by 4.1 percent and within-school test-score dispersion by 25%. Over the average closure of about 53 days, students from low-income households experience approximately a 0.09 SD decline in test scores. To understand substitution between in-person schooling and parental investment and to assess a compensatory policy, we build and estimate a static tournament model in which formal schooling and tutoring jointly generate test scores. Estimates suggest that tutoring substitutes for in-person school days and that closures burden low-income households disproportionately. Counterfactual simulations show that a means-tested tutoring voucher can fully offset the increase in test-score inequality caused by school closures.

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Keywords: school closure, parental investment, education inequality, learning loss, education voucher

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1. Introduction

School closures increase educational inequality across socioeconomic groups, whether caused by forced displacement (Kim 2024), natural disasters (Sacerdote 2012; Andrabi et al. 2023), or a global pandemic (Agostinelli et al. 2022; Jack et al. 2023; The Economist 2021). Households may compensate for lost in-person schooling with market substitutes such as private tutoring, but students differ in how much they benefit from in-person schooling and in their access to these substitutes, which can widen achievement gaps. While recent studies have used economic models to examine school closures’ impact on educational outcomes (Grewenig et al. 2021; Agostinelli et al. 2022; Fuchs-Schündeln et al. 2022), evidence remains limited on how closures generate different parental responses across socioeconomic groups, how these responses contribute to educational inequality, and which interventions can mitigate the increased inequality due to school closures.

In this paper, we address this gap in two steps. First, we empirically identify the impact of additional school closure days on private tutoring expenditure and academic achievement, using a novel administrative-linked panel dataset from South Korea.¹ Second, we build and estimate a tournament model of parental investment in which in-person school days enter the education production function as an input. We then use the estimated model to assess learning losses among low-income households, quantify how closures affect educational inequality, and evaluate the size of a tutoring voucher that could offset the resulting increase in inequality.

We choose South Korea and its response to the COVID-19 school closure as a laboratory to facilitate causal interpretation. The country experienced extensive school closures, with only 6.2% of school days being fully in-person during the first two years of

¹We use the terminology “private tutoring” to encompass various forms, including one-on-one private tutoring, group private tutoring, and hagwon (private tutoring institutions in South Korea). In the education literature, this broad category is often referred to as “shadow education” (Bray 1999). We discuss the supply side of tutoring in Section 2.2.

the pandemic.² Schools were fully closed for 24.7% of school days, and partially closed for 67.6% of days.³ The plausibly exogenous nature of school closures reduces concerns about reverse causality. The country has been successful in “flattening the curve” with no blanket lockdown. This success makes our estimates less contaminated by the aggregate environment, compared to other developed countries. Moreover, near-universal school attendance and uniform pre-pandemic in-person school days make treatment intensity unlikely to reflect pre-existing differences in attendance behavior across schools or households.⁴

We construct a novel student panel by linking survey and administrative sources. The survey sources are the Gyeonggi Educational Panel Survey (GEPS) and the Busan Educational Longitudinal Survey (BELS), two regionally representative longitudinal surveys of South Korean students. We combine them with two administrative sources: SCHOOLINFO, a census of Korean secondary schools, and the National Education Information System (NEIS), which provides the universe of registered hagwons (cram schools). The resulting dataset follows students from multiple adjacent cohorts and combines student-level information on private tutoring expenditure, nationally standardized test scores, and socioeconomic background with school-level in-person school days and neighborhood-level hagwon supply. This structure allows us to track within-student changes, exploit quasi-random variation in school-closure intensity, and compare COVID-affected cohorts with pre-pandemic cohorts.

We employ a continuous difference-in-differences design that exploits policy-driven, quasi-random variation in closure days across schools. At the individual level, we find that 10 additional closure days lead to a 4.1% increase in private tutoring expenditure.

²South Korea recorded the highest school closure ratio among Asia & Pacific countries in the World Bank high-income group.

³We define partial closure as having at least one but fewer than 95 full closure days during the spring semester. The detailed distribution is reported in Section 2.3.

⁴In the 2019 spring semester, school days had a mean of 95.8 and a standard deviation of 2.0. South Korea has the OECD’s lowest absenteeism rates: in PISA 2018, about 2% of Korean 15-year-olds reported skipping a full school day in the two weeks before the test, versus an OECD average of 21% (OECD 2019).

We also find that 10 additional closure days reduce test scores by 0.02 SD for low-income students, equivalent to about 0.09 SD over the average closure of 53 days, with no significant effect for middle- or high-income students. Consistent with these heterogeneous impacts, the within-school standard deviation of test scores increases by approximately 25 percent per 10-day loss in in-person days, while there is little impact on the within-school dispersion of private tutoring expenditure. These findings suggest that while households compensate for lost school days through private tutoring, this compensation is uneven across socioeconomic groups, potentially widening educational inequality.

Motivated by our quasi-experimental findings, we develop a structural model that captures two distinct channels of heterogeneity, households' financial capacity to substitute toward private tutoring and the direct return to in-person instruction. We use the structural model to address two questions: how does each channel contribute to the changes in the unconditional dispersion of test scores? And, when tutoring support is offered, how large a targeted voucher is sufficient to reduce inequality? We adopt a tournament framework of Kang (2024), in which parents choose private tutoring to maximize expected utility. This model formalizes parents' incentives to invest in their children's education, the joint role of private tutoring and in-person school days in producing test scores, and the extent to which these two inputs substitute for one another. A college-admission tournament sets rewards by rank, while a constant elasticity of substitution (CES) production function combines previous test score, school days, and tutoring. This specification, combined with the quasi-experimental variation in school closures, allows us to identify both the marginal effect of each input and the substitution elasticity between the number of school days and private tutoring expenditure.⁵

We estimate the model by maximum likelihood with a moments penalty, requiring the simulated data to match our two-way fixed effects (TWFE) estimates for mean tutoring and within-school score dispersion, while also satisfying the counterfactual parallel-trend

⁵An alternative model assuming individuals care about absolute performance (the "Max-Score" model) substantially overpredicts parental investment levels.

restriction. This procedure disciplines structural estimation by both the reduced-form treatment effects and the identifying restriction on which they rely.

Our structural estimates imply that private tutoring and school days are gross substitutes. Based on the CES specification, substitutability implies that the marginal effects of school days are higher for low SES households. This finding aligns with our TWFE results showing stronger impact of school closures on low-SES households.

We use the estimated model to conduct counterfactual experiments on private tutoring voucher policies. First, we quantify that restoring pre-COVID school days would reduce within-school score inequality by approximately 8% while only modestly affecting mean performance. We use this full-reopening scenario as a benchmark for the educational losses associated with school closures. We then simulate voucher policies that vary in voucher size and eligibility criteria. Voucher size has diminishing returns in reducing inequality. Beyond a threshold, larger vouchers exacerbate inequality. Consistent with the heterogeneous effects observed during school closures, our analysis shows that targeted voucher policies could fully offset the increased inequalities induced by school closures. Specifically, a voucher of approximately 250,000 KRW, or about \$210, targeted at the bottom 60% of the income distribution restores within-school test-score dispersion to the level that would prevail under full school reopening. Our structural analysis emphasizes the equalizing role of public education and suggests that targeted voucher policies can mitigate test score inequality when full school reopening is not feasible.

Our paper contributes to several strands of literature. First, we provide causal estimates of the impact of school closures on parental investments and educational inequality, extending the literature on educational disruption. Previous work has documented learning losses after shocks such as natural disasters (Sacerdote 2012; Andrabi et al. 2023), school displacement (Kim 2024), and pandemic-era school shutdowns (Bacher-Hicks et al. 2021; Maldonado and De Witte 2022; Jack et al. 2023; Hahn et al. 2023). These studies

consistently find declines in average test scores and increasing test score disparities, but do not investigate parental behavioral responses. We leverage quasi-experimental variation in school closures and the longitudinal dataset with individual fixed effects to quantify how households differentially adjust educational investments when faced with school disruptions.

Second, our paper contributes to the literature on ex-ante policy evaluation aimed at mitigating educational inequalities when public education is disrupted. Our production function estimates embedded in the structural model imply that private household investments and public educational inputs are substitutes. Our substitutability finding also adds to a longstanding literature on how parents reallocate own-child investment in response to public inputs (Becker and Tomes 1986; Todd and Wolpin 2003; Datar and Mason 2008; Das et al. 2013; Fredriksson et al. 2013; Greaves et al. 2023). We use the structural model to assess the effectiveness of a targeted tutoring voucher policy in offsetting learning losses. Closest to our approach, Agostinelli et al. (2022) show that low-income children lost substantially more ground during COVID-19 because affluent households could better compensate for lost schooling, and discuss potential policy interventions. In line with their approach, we construct a structural model to map households' heterogeneous ability to replace lost instructional time, investigating mechanisms through which school closures magnify educational inequality and evaluating policy tools to mitigate these effects.

Third, our estimation approach contributes to recent empirical work integrating structural models with experimental identification (Heckman 2010; Galiani and Pantano 2021; Todd and Wolpin 2023), and in particular, quasi-experimental variation (Busso et al. 2013; Voena 2015; Blundell et al. 2016; D'Haultfœuille and Février 2020; Alves et al. 2023). Building on this line of research, we formally incorporate the parallel-trends assumption from our continuous difference-in-differences design into the maximum

likelihood estimation of the structural model. By doing so, our method ensures internal consistency between reduced-form evidence and model-based counterfactual analysis.

The rest of the paper is organized as follows. Section 2 outlines the background and data. Section 3 describes our empirical framework, followed by our results in Section 4. Section 5 defines the tournament model. Section 6 describes how we implement the structural estimation and reports our results. Section 7 conducts counterfactual policy experiments based on the estimated structural model.

2. Institutional Background and Data

2.1 Background on the Korean Public Education

South Korea operates a highly centralized education system with near-universal enrollment and a standardized 6-3-3 structure: six years of elementary school, three years of middle school, and three years of high school. The government requires 190 school days annually and mandates adherence to the National Curriculum for Primary and Secondary Schools.⁶ Local school boards exercise only limited authority in educational policy. Parental involvement plays a minimal role in school governance compared to countries like the United States. Opting out of the formal school system is virtually impossible, with homeschooling both legally prohibited and rarely practiced.

This centralized system reflects the country's famous 'education fever' – an exceptional cultural emphasis on schooling as the primary engine of economic growth and social mobility. This national obsession translates into consistent educational participation, with South Korea maintaining the OECD's lowest absenteeism rates (OECD 2019). The government's commitment to educational continuity has been remarkable—prior to COVID-19, schools remained open through multiple national crises, including military

⁶It applies to both public and private schools. A few selective high schools (autonomous private schools and special purpose schools) may customize the curriculum.

coups (1961, 1979) and financial crises (1997, 2007). Notably, the government made no exceptions for previous epidemics that threatened public health, such as MERS (2015), H1N1 (2009), SARS (2004), and even the 1969 cholera outbreak – which posed a serious health challenge during a period when the country was still economically underdeveloped.

This centralized institutional structure also governs school assignment. Secondary school assignment is largely random conditional on the residential school district. Students entering middle or general high school are assigned by lottery to one of the several schools that together cover their district, which typically spans a radius of roughly 5 to 8km. Residential sorting rarely happens once children enter secondary school, and the within-district lottery randomizes school assignment among neighborhood applicants. These features produce an unusually homogeneous secondary schooling environment (Park et al. 2013, 2018).

2.2 Private Tutoring and Hagwon (Cram School)

In this relatively homogeneous public school system, private tutoring becomes a central margin of test-score competition. There exists an extensive private tutoring market. ‘Hagwon’ (cram schools) provide supplementary tutoring services widely utilized across all school levels.⁷ These services primarily target college admission preparation, which directly impacts lifetime earnings prospects. Although private tutoring is broadly accessible, quality varies substantially with cost, earning it the label of “Great Unequalizer.” The government has repeatedly attempted to regulate this market, most notably through business-hour restrictions, which were challenged in court but upheld by the Constitutional Court. Despite these regulatory efforts, the private tutoring market continues

⁷According to Ministry of Education data (2010), 45% of elementary, 50% of middle school, and 84% of high school students participate in private tutoring services.

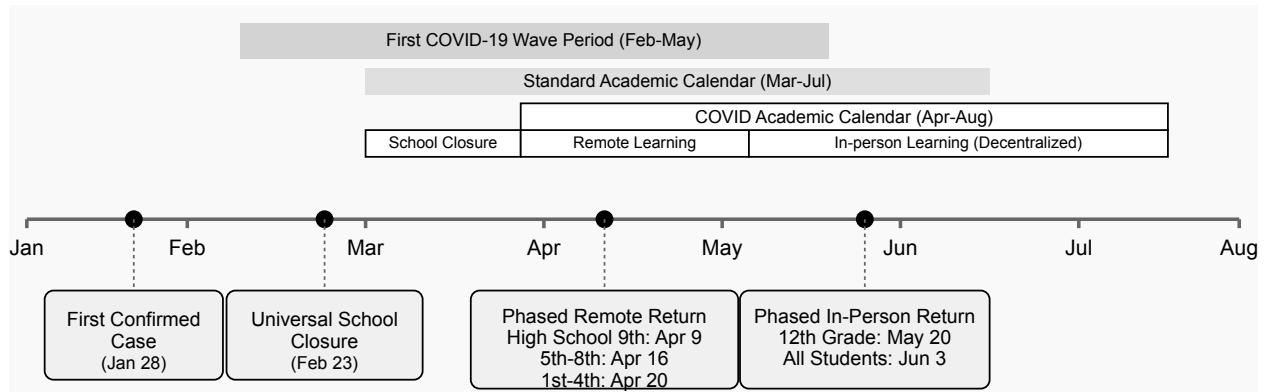
to thrive, offering students a variety of options to fuel their education fever beyond the formal system.

The Korean private tutoring market is heterogeneous in both form and scale. Households can purchase tutoring through several channels: private cram schools (*hagwons*), individualized one-on-one tutoring, online subscription platforms (e.g., Megastudy, Etoos), and school-based after-hours programs. Among these, hagwons account for the largest share of household spending and student enrollment, particularly at the middle and high school levels where college-admission preparation dominates demand. The supply of hagwons is also thick relative to the student population. In January 2020, the Ministry of Education's National Education Information System recorded 27,966 registered hagwons in Gyeonggi province, or about 19 per 1,000 elementary, middle, and high school students, and 6,855 in Busan, or about 23 per 1,000 students. Providers are concentrated around residential and school neighborhoods and are broadly co-located with the student population, suggesting limited access frictions in choosing among providers.

2.3 The COVID Responses in Early 2020

The COVID-19 pandemic brought an unprecedented challenge to South Korea's education system, triggering the first nationwide school closure in the country's history. When the first case was reported in January 2020, the Ministry of Education maintained the normal academic calendar, which starts on March 2nd. However, as the first wave arrived in mid-February, the Ministry declared the first-ever nationwide school closure on February 23rd, postponing the March 2nd school opening. This "temporary" closure extended five weeks, transitioning to a fully remote opening with a phased schedule: grades 9-12 returned April 9th, grades 5-8 on April 16th, and grades 1-4 on April 20th. Figure 1 illustrates this timeline of closures and phased reopenings.

As the first wave subsided, the Ministry planned a return to in-person schooling. A phased return began with 12th graders on May 20, with all students back by June 3. This

Figure 1: South Korea's COVID-19 Education Response Timeline (Early 2020)

marked the end of nationwide closures, which did not recur. Instead, the Ministry issued guidelines but delegated open-or-remote decisions to regional education offices and school principals. This decentralized approach generated the school-level variation in in-person school days that underpins our identification strategy. Figure A.2 shows the spatial variation in the regions in our dataset. In our sample, the spring 2020 in-person school days have a mean of 42.5 days and a standard deviation of 7.1 days, against a pre-pandemic full-open baseline of 95 days (95.8 days in 2019, with a standard deviation of 2.0). About 96.1% of treated students attended schools delivering between 30 and 70 in-person days. The minimum was 20 days, with no school falling below that floor. Only 0.2% of students attended schools at or near the pre-pandemic baseline (≥ 85 days).

While schools closed, private tutoring institutions remained open throughout this period. South Korea implemented its ‘Trace, Test, and Treat (3T)’ strategy rather than imposing blanket lockdowns.⁸ This policy regime made hagwon-specific closures infeasible, leaving hagwons subject mainly to social-distancing measures no more stringent than those applied to other businesses. Online private tutoring services operated with even fewer limitations, as they were not constrained by physical distancing requirements. Meanwhile, with parents still required to work in-person due to the absence of workplace

⁸See Bicker (2020) for details.

lockdowns, many households had limited ability to supervise children’s education at home, making private tutoring their only viable option. In effect, the pandemic shut down the ‘Great Equalizer’ (public schools) while unleashing the ‘Great Unequalizer’.

2.4 Data

This section describes our novel panel dataset, which links student-level longitudinal surveys with two administrative sources: a census of secondary schools and a registry of private tutoring providers geocoded to each school’s neighborhood.

Our analyses use three cohort groups from two regionally representative longitudinal surveys of South Korean students. Each cohort is defined by their grade level during the COVID-19 pandemic in 2020: 8th graders, 11th graders, and 12th graders. The 8th- and 11th-grade cohorts come from Busan, one of Korea’s largest metropolitan cities, and the 12th-grade cohort from Gyeonggi, its most populous province.⁹ These datasets provide rich information at student, parent, and school levels, including test scores, private education expenditure, and school budgets. We link these surveys with SCHOOLINFO, an administrative dataset containing in-person school days for all primary and secondary schools.

Our identification strategy leverages in-person school day variations within the COVID group, using the Control group for comparison, as we discuss in Section 3 in detail. We combine the cohorts who experienced COVID school closures to form our treatment group (also referred to as the COVID group). Each survey includes multiple cohort waves, allowing us to construct a corresponding control group of students who attended the same grade levels before the pandemic: 8th graders from 2017, 11th graders from 2013, and 12th graders from 2014. These control cohorts, unexposed to the pan-

⁹These data come from the Gyeonggi Educational Panel Survey (GEPS) and the Busan Education Longitudinal Survey (BELS). Both surveys share a common design based on the Korean Education Longitudinal Study (KELS), which adapted the Education Longitudinal Study (ELS) of the United States.

demic, provide baseline information on student behavior and learning outcomes in the absence of school closures.¹⁰ Table 1 summarizes our data structure.

Table 1: Definition of COVID and Control groups

Survey Year(s)	8th graders		11th graders		12th graders	
	COVID	Control	COVID	Control	COVID	Control
2012						10th
2013						11th
2014						12th
2015						
2016		7th		10th		
2017		8th		11th		
2018	6th		9th		10th	
2019	7th		10th		11th	
2020	8th		11th		12th	

We focus on parental investment and student academic achievement. Our main measure of parental investment is private tutoring expenditure.¹¹ In both surveys, parents report monthly spending (in Korean won) on private tutoring across three core subjects: Korean language, Mathematics, and English. Since the first semester of the Korean academic year runs from March to July, with surveys administered between late July and mid-August, the response effectively captures private tutoring during the spring semester. We take a sum of expenditures across all three subjects as our main measure of parental investment. Throughout the analysis we apply the inverse hyperbolic sine (IHS) transformation to tutoring expenditure and household income (Mullahy and Norton 2024).¹²

¹⁰Due to the survey timing, the “-2” years’ information of 8th and 11th graders is unavailable for the Control group.

¹¹We do not consider time investment of parents, as it becomes less relevant once students enter secondary school in South Korea (See Section 3 of (Kang 2024)) .

¹²In particular, $\text{arcsinh}(y) = \ln(y + \sqrt{y^2 + 1})$, which is defined at zero, accommodating the approximately 5% of households with no tutoring expenditure. For positive values it closely approximates the natural log, so regression coefficients retain an approximate percentage-change interpretation.

For academic achievement, we use scores from national-level standardized tests, which allow for valid comparisons of student performance across different schools and regions. Students took the exam at the end of the semester.¹³ The raw test scores range from 0 to 100 for each subject: Korean, English, and Mathematics. These scores are standardized to have a mean of 100 and standard deviation of 20 within each cohort and grade. We use the average of these standardized scores across all three subjects as our main measure of academic achievement.¹⁴¹⁵ Table A.1 presents the descriptive statistics of the working sample.

SCHOOLINFO provides data on total in-person school days for all primary and secondary schools by school and semester for 2019 and 2020. We link these figures from spring semesters to the schools in our sample. Consequently, while our parental investment and test score measures exist at the individual student level, in-person school days are measured at the school level. This variable is not included in the dataset for years prior to 2019.¹⁶ In any case, in-person school days varied little across schools in the pre-pandemic period under the centralized system. For earlier years, we assume all students attended 95 days, corresponding to a full pre-pandemic spring semester. We choose the spring-semester baseline because the COVID-induced school closure occurred during spring 2020, and to align the treated cohort with the control cohorts within the same spring-semester window. The observed 2019 pre-COVID data support

¹³Although earlier national tests during the 2020 academic year were disrupted (the March test was cancelled and the April test was postponed to May), our analysis uses the June test, which was administered as scheduled.

¹⁴For 12th graders in the COVID group, scores are provided in stanine format. We map each stanine bin to original scores using external data sources that provide the cut points for this exam.

¹⁵In reduced-form regressions (Sections 3 and 4) the outcome is rescaled to within-cohort-grade standard-deviation units (mean 0, standard deviation 1), so coefficients are reported in SD units. The structural model in Section 5 retains the native mean-100, SD-20 scale, and the within-school standard deviation in Table 6 and Table 8 is reported on that scale (e.g., a closure-days coefficient of 0.270 corresponds to roughly 25 percent of the baseline within-school standard deviation in raw score points). The 25-percent figure in the abstract and introduction expresses the rise in within-school dispersion in these proportional terms.

¹⁶SCHOOLINFO has routinely collected attendance data as part of its administrative records, but research access has been restricted. The COVID-19 pandemic created exceptional circumstances that allowed researchers to access attendance data from 2019 onward. Nevertheless, pre-2019 records remain unavailable for research use.

this assumption: in-person school days varied little across schools, with a mean of 95.8 days and a standard deviation of 2.0.

Finally, we measure the local supply of private tutoring using administrative records from the National Education Information System (NEIS), which lists all registered hagwons with their business name and address. We geocode each address via the Kakao Map API and, for each school-year, construct counts of registered hagwons within 1km, 3km, and 5km of the school, measured as of January 1. The 3km count is our preferred measure.¹⁷

3. Empirical Strategy

We estimate the impact of school closures on parental investment and test scores. The source of identification is the decentralized variation at the school level in the number of closure days, conditional on the student fixed effects. In addition, we use the pre-COVID cohorts as a control group to adjust for grade fixed effects that capture grade-to-grade growth in outcomes. We estimate the following two-way fixed effect model:

$$Y_{ikg}^c = \beta_0 + \beta_1 \text{closures}_{kg}^c + \Gamma Z_{ikg}^c + \mu_i + \eta_g + \varepsilon_{ikg}^c. \quad (1)$$

In the above equation, Y_{ikg}^c is an outcome variable for student i in school k , grade g , and cohort c . The primary outcomes we examine are private tutoring expenditure

¹⁷The choice of radii rests on two assumptions. First, we assume that students' one-way commuting distance to hagwons does not differ substantially from that to school, given that both destinations serve the same student population within the same residential geography. This assumption is not directly testable with our data, but we regard it as reasonable in the Korean context where hagwons are predominantly located near schools and residential clusters. Second, we assume that students' commuting distance does not exceed 10km, which we validate using individual-level commuting data from the 2015 Census (2% sample); we prefer this wave over the 2020 Census to avoid contamination from pandemic-era behavioral changes. Among secondary school students, 69.4% commute within their town of residence (eup/myeon/dong), with an average radius of approximately 3 km; the remaining 30% commute to an adjacent town, typically when the assigned school lies closer to a town boundary than to the school within their own town. Average one-way commute time is 22.5 minutes, with roughly half of students traveling on foot and 40% by bus or metro, consistent with a 10km upper bound.

(IHS-transformed) and standardized test scores. We denote $g = 0$ as the reference grade in which COVID cohorts experienced the pandemic-induced school closures. All other grades are indexed relative to this reference point across both treatment and control groups. For instance, $g = 0$ corresponds to the 2020 academic year for 8th graders in the COVID group, whereas it represents 2017 for their counterparts in the control group. The variable $closures_{kg}^c$ denotes the number of school closure days, measured in tens of days.¹⁸ The vector Z_{ikg}^c is a set of time-varying individual-level controls. We control for an individual fixed effect μ_i and a grade fixed effect η_g . We also conduct analyses by subgroup based on student and parent characteristics.

This is a standard continuous difference-in-differences design where we use the cumulative number of school closure days during Spring 2020 ($g = 0$) as our treatment intensity. This treatment affects only the COVID cohort, with different intensities across schools, and our data directly provide school-level counts of in-person days rather than day-by-day closure records. Since we have only this single treatment period, the post-treatment period dummy is omitted. Unlike staggered adoption designs with varying treatment timing across units, our approach examines treatment intensity within a uniform timeframe. Our identification relies on the key assumption that without COVID-induced school closures, outcome trends would have evolved in parallel across the COVID and Control groups, conditional on controls and fixed effects. The validity of these assumptions is discussed in detail below.¹⁹

¹⁸Specifically, $closures_{kg}^c \equiv \bar{s} - schdays_{kg}^c$, where $\bar{s} = 95$ is the minimum attendance requirement per semester and $schdays_{kg}^c$ is the number of in-person school days. Relative to an equivalent specification in $schdays_{kg}^c$, the signs of β_1 in Equations (1) and (2) and of both β_1 and β_2 in Equation (3) are reversed; all other coefficients are unaffected.

¹⁹An alternative would be to instrument tutoring expenditure with a pandemic-induced shock. We consider two candidate instruments. The first is local COVID-19 incidence at the si/gun/gu level; in our sample regions (Gyeonggi and Busan), case counts during spring 2020 were small, the first stage is weak, and the IV/2SLS estimates are uninformative (Appendix Table B.1). The second is school closure days, which has a much stronger first stage but plausibly affects test scores directly via the loss of in-person instruction; the resulting IV/2SLS estimates are statistically insignificant in the second stage (Appendix Table B.2). We therefore use our structural model in Section 5 to disentangle the direct effect of school closures from the indirect effect operating through tutoring.

We also examine whether school closures exacerbated educational inequality in dimensions not fully captured by changes in individual means. To address this question, we estimate a school-level model analogous to Equation (1):

$$Y_{kg}^c = \beta_0 + \beta_1 \text{closures}_{kg}^c + \Gamma Z_{kg}^c + \mu_k + \eta_g + \varepsilon_{kg}^c. \quad (2)$$

Here, Y_{kg}^c represents the standard deviation of test scores within school k . This measure allows us to assess how school closures affected the dispersion of academic performance, given our premise that public education serves as a “Great Equalizer.”

For both Equations (1) and (2), the coefficient of interest β_1 captures the change in the outcome variable due to a ten-day increase in school closure days. We report standard errors clustered at the school level for our individual-level regressions.²⁰

We discuss our four identifying assumptions and two supporting facts below. The validity of our empirical approach rests on these.

Assumption 1 *Variation in school closure days is random, conditional on time-invariant individual characteristics.*

The key variation for identification is the number of school closure days, determined by school principals and the Regional Offices of Education, as discussed in Section 2.3. No individual student could predict or affect the extent of school closures. One of the major complaints from students and parents during this period was the uncertainty about whether schools would open the following week or, in extreme cases, the very next day.

In principle, reverse causality could arise if reopening decisions were affected by households or correlated with components of the individual-level error term.²¹ However,

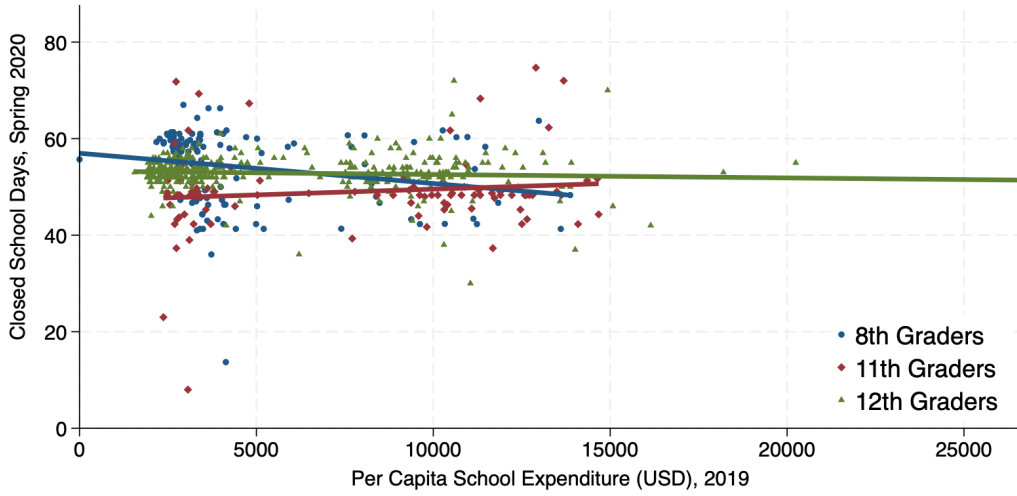
²⁰For the small-sample school-level balance and dispersion-by-income checks reported in the appendix, we cluster at the city level.

²¹Parolin and Lee (2021) reports that school closures from September to December 2020, in the United States, were more common in schools with higher shares of students from racial and ethnic minorities, who are likely to have poor socioeconomic status.

this is unlikely in our context for two reasons. First, although the opening decision was decentralized, it remained in the hands of policymakers familiar with the top-down approach, not with parents or students. Second, potential channels for reverse causality, such as selective schools having different incentives for reopening, appear absent in the data. For instance, selective schools might prefer reopening to provide intensive in-person instruction, or conversely, might remain closed if their students can effectively access private tutoring.²² To investigate this possibility, we first examined whether school quality, proxied by per capita spending, predicts the number of school closure days (Figure 2). There are no systematic correlations, and conditioning on school characteristics such as public-private status does not change the result. Secondly, we verify that school closure days are uncorrelated with school-level socioeconomic composition: regressing closure days on average log household income and mother’s education yields economically negligible and statistically insignificant coefficients (Appendix Table B.3). Finally, we find that the number of cram schools within the school neighborhood is not correlated with the school-closures decisions.

There could be measurement error due to COVID infection. If a student were infected, there would have been additional “closure” days at the individual level, beyond the school-level policy captured by $closures_{kg}^c$. This would constitute nonclassical measurement error since it correlates with the idiosyncratic error term: sick students cannot participate in private tutoring. While this type of measurement error is a valid concern for COVID-era data, it is also unlikely in our context. First, the number of school-age cases was limited in our sample provinces, Gyeonggi and Busan. As of July 31st, 2020, there were 80 out of 1,537 total cases in Gyeonggi (0.07% of the teenage population). For Busan, while we do not have the number of cases by age, there were only 168 cases in total by July 31st, 2020 (0.05% of the population)—this represents the upper bound of school-age cases in

²²For non-Korean readers, the second scenario considers the following story. Suppose that the objective function of a school is to maximize the number of students who gain admission to top colleges. Private tutoring is generally more “effective” for preparing for exams. In this case, schools have incentives to close in order to let students maximize their private tutoring hours.

Figure 2: School Closure Days and Per-capita School Expenditure

Data: GEPS, BELS, and SCHOOLINFO. The per capita school expenditure is converted from Korean Won (KRW) to US Dollar (USD) assuming 1 USD = 1,200 KRW.

Busan. Furthermore, we found no association from a regression of closure days on the number of cumulative cases in subregions of the two provinces.

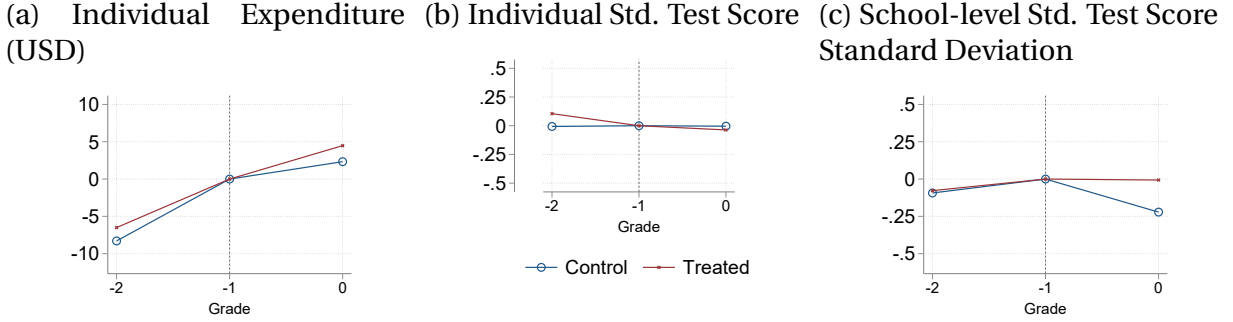
Assumption 2 *Parallel trends hold for parental investment, test scores, and the school-level standard deviation of test scores.*

The COVID shock is an aggregate shock that absorbs into the time dimension. In order to identify the COVID effect, we assume that the annual growth of private tutoring expenditure is the same across the treated and control cohorts. That is, the growth due to the grade effect is time invariant. This parallel trends assumption enables us to use an earlier cohort in the dataset as a control group, attributing all 2020-specific outcome variations to the changes in school closure days. Figure 3 visualizes the average tutoring expenditure growth of the treated and control cohorts between $g = -1$ and $g = 0$.

Assumption 3 *The individual fixed effects absorb most unobserved heterogeneities, and bias due to time-varying confounders is limited.*

First, we follow the standard two-way fixed effects framework. The individual fixed effects control for all time-invariant characteristics at the student level, including demographics,

Figure 3: Trends of Outcome Variables



Note: All outcomes are normalized to have zero mean by sample group at $g = -1$. The test score plots' vertical axis is measured in standard-deviation units.

household background, and baseline academic performance. The grade fixed effects account for outcome growth over time that is common across all students.

As part of Z_{ikg}^c , we control for time-varying measures such as household income in expenditure regressions and expenditure in test score regressions. Given our short data timeframe, few other relevant individual controls vary within this period. Our analysis confirms that results remain robust when including these controls, suggesting minimal bias from time-varying confounders.²³

Assumption 4 *Treatment effects are homogeneous across treatment intensities.*

Our continuous difference-in-differences design requires that the effect of an additional ten days of school closure is similar across the distribution of treatment intensity. This allows us to interpret our estimated coefficient as the average treatment effect per ten days. While recent literature highlights concerns with heterogeneous treatment effects in difference-in-differences models (Callaway et al. 2024), our non-staggered treatment

²³Including time-varying controls in two-way fixed effects models requires strong identification assumptions that treatment should not affect such controls (Caetano et al. 2024). As we discuss in Fact 1 below, since South Korea avoided economywide lockdowns and experienced limited labor market disruption, household income likely remained relatively stable during our period of interest. Expenditure, however, did respond, and it is one of our main findings. We nonetheless include expenditure in test score regressions because of its relevance, while acknowledging this potential endogeneity. Our main estimates are robust to including these controls.

setup avoids many of these concerns. To assess the remaining concern empirically, we estimate a modified version of Equation (1):

$$Y_{ikg}^c = \beta_0 + \beta_1 \text{closures}_{kg}^c + \beta_2 \text{closures}_{kg}^c \times HIGH_i + \Gamma Z_{ikg}^c + \mu_i + \eta_g + \varepsilon_{ikg}^c. \quad (3)$$

where $HIGH_i$ is a dummy equal to 1 if closures_{kg}^c is above-median in 2020.²⁴ The coefficient β_2 captures any differential effect between above- and below-median treatment intensity groups. We test the school-level treatment as well, by similarly augmenting Equation (2). The results of these tests are discussed in our robustness checks section.

In addition to our four identifying assumptions, we leverage the following empirical facts to argue that South Korea is one of the best “laboratories” for identifying the impact of COVID-induced school closures, and our estimates are less prone to aggregate-level confounders compared to those from other developed economies.

Fact 1 *Without economywide lockdowns, South Korea’s economy underwent minimal disruption during early COVID-19, compared with other developed economies.*

While COVID-induced school closures in South Korea offer quasi-random identifying variation, isolating their effects from broader pandemic impacts requires addressing potential confounding from aggregate COVID effects. In most developed economies, parental job loss and school closures are deeply intertwined, confounding their independent effects. South Korea provides a unique research environment not only for its quasi-random school closure variation but also for its successful ‘flattening the curve’ strategy that avoided economywide lockdowns while implementing targeted social distancing, testing, and financial stimulus. Consequently, employment rates remained stable, deviating only by 1-2 percentage points from pre-pandemic trends (Figure A.3), whereas U.S. employment fell by 10 percentage points between February and April

²⁴Using $HIGH_c$ and $HIGH_i$ yields identical results as closures_{kg}^c varies at the school level.

2020.²⁵ That is, the universal school closure was a preventative measure rather than a response to widespread outbreak.²⁶ This relative economic stability allows changes in parental investment to be more confidently attributed to school closures rather than labor market shocks.

Fact 2 *School attendance is universal in South Korea, and its private tutoring market is robust.*

Our research design benefits from both universal treatment compliance and wide access to educational alternatives, which are characteristic of the country's 'education fever.' South Korea's formal education system exhibits exceptionally high attendance rates, with chronic absenteeism rarely observed. It implies that when schools closed, virtually the entire student population was affected, creating a treatment group with minimal selection concerns.

Simultaneously, South Korea's private tutoring market remained operational throughout our periods of interest. Administrative records from the National Education Information System (NEIS) confirm that the supply of hagwons did not contract during the pandemic in our sample regions (documented in Section 4). The market offers a comprehensive range of options from free online lectures to premium one-on-one tutoring services. These alternatives made substitution away from public schooling feasible and observable during school closures. Because approximately 75% of students already used some form of private tutoring before the pandemic, many households could adjust their existing tutoring arrangements in response to school closures.

In sum, these assumptions and institutional features provide a research design that isolates the effect of school closures from broader pandemic disruptions. Our approach

²⁵Using weekly administrative payroll data, Cajner et al. (2020) reported that U.S. employment fell by 21% through late-April 2020, and 30 percent of the decline was driven by business shutdowns.

²⁶While the first wave of infections in February triggered massive policy responses, the labor market shock was limited to Daegu, the region hit by the wave (Aum et al. 2021). Our sample does not include Daegu. See Figure A.4 for employment rates by provinces.

leverages South Korea’s unique combination of quasi-random school reopening decisions, stable economic conditions, and accessible educational alternatives. This setting enables us to examine not only the average impact of reduced in-person schooling days on individual outcomes but also its distributional consequences for educational inequality.

4. Empirical Results

We begin by discussing our baseline results, the effect of school closures on individual-level outcomes. Next, we report our heterogeneity analysis for various student subgroups. We then examine school-level outcomes to assess how school closures affected educational inequality.

4.1 Individual-Level Effects

Table 2 presents our individual-level results based on Equation (1) using IHS-transformed private tutoring expenditure and standardized test scores. For each outcome, we present estimates without time-varying covariates (columns 1 and 6), and with covariates (columns 2 and 7). We also include tests for treatment heterogeneity using Equation (3) (columns 3 and 8). Columns 4–5 and 9–10 add hagwon density controls, discussed in the robustness paragraph below. We consider columns (2) and (7) our preferred specifications.

For private tutoring expenditure, we find that ten additional school closure days raised investment by approximately 4.1%, statistically significant at the 1% level. This estimate remains stable when controlling for household income (column 2), addressing concerns about time-varying confounders (Assumption 3). The interaction term with above-median closure days $\hat{\beta}_2$ in column (3) is tiny and insignificant, indicating that heterogeneity in treatment effects across schools with different levels of closure intensity is of less concern (Assumption 4). Appendix Table B.4 reports the same estimates in

absolute levels (10,000 KRW), an overall increase of approximately 5,430 KRW per ten closure days.

For test scores, we find a precisely estimated zero effect on average across all specifications (column 6).²⁷ It implies that students in schools with more closure days did not score systematically higher or lower than those with fewer closure days, conditional on their characteristics. The null is robust to controlling for private tutoring expenditure (column 7), and the treatment heterogeneity interaction is small and insignificant (column 8).²⁸

Table 2: Impact of School Closure Days on Individual Outcomes

	IHS(Parental Expenditure)					Test Score				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Closed School Days	0.040*** (0.005)	0.041*** (0.005)	0.041*** (0.006)	0.041*** (0.005)	0.042*** (0.005)	-0.005 (0.005)	-0.006 (0.005)	-0.004 (0.005)	-0.006 (0.005)	-0.006 (0.005)
IHS(income)		0.215*** (0.051)	0.215*** (0.051)	0.214*** (0.051)	0.214*** (0.051)					
IHS(expenditure)							0.024*** (0.006)	0.024*** (0.006)	0.024*** (0.006)	0.024*** (0.006)
Above Median $\times$ Closed School Days			-0.002 (0.006)					-0.005 (0.005)		
log(Hagwon counts)				0.022 (0.023)	-0.010 (0.029)				0.008 (0.021)	0.004 (0.023)
log(Hagwon counts) $\times$ Closed School Days					-0.006* (0.003)					-0.001 (0.003)
N	17,104	17,104	17,104	17,104	17,104	17,104	17,104	17,104	17,104	17,104

Note: Clustered standard errors in parentheses (school level). ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

These findings jointly show that households increased private tutoring investment while the average test score gradient across closure intensities remained flat. Appendix Tables B.5 and B.6 show how our estimates change across specifications. We find estimates from the COVID-sample alone differ from our full-sample results, which shows the importance of our identification strategy in isolating school closure effects from confounding grade-specific trends and the aggregate COVID effect.

²⁷Because scores are standardized within cohort-grade cells, the score regressions are identified from cross-school variation in closure days within those cells.

²⁸Using different data, Hahn et al. (2023) also find near-zero average test score effects using a nationally representative sample of Korean students.

Robustness to Supply-side Changes Provider-driven changes, such as price adjustments, could partly account for the results. We address these concerns directly as follows. First, using administrative data from the National Education Information System (NEIS), we document that the number of registered hagwons did not decline during the pandemic in either sample region: the count in Gyeonggi province was virtually unchanged (27,966 in January 2020 vs. 28,021 in January 2021), and the Busan count slightly rose from 6,855 to 7,755. Moreover, school closure days and local hagwon counts are uncorrelated. Second, we add the log count of registered hagwons as a control and interact it with school closure, finding that our main expenditure estimate remains unchanged as shown in columns (4) and (5) of Table 2. Third, using the subsample of 12th graders for whom weekly tutoring hours are observed, we decompose expenditure into tutoring *quantity* (hours) and *quality* (IHS expenditure per hour). Ten additional closure days increase weekly tutoring time by approximately 2 hours but modestly lower per-hour expenditure by 2.7% (Appendix Table B.7). This pattern holds across all income terciles, providing little evidence that provider-driven price changes explain the larger percentage expenditure response among lower-income households (Appendix Table B.8). Together, these results show that the expenditure response reflects adjustments in tutoring hours rather than higher prices per unit.

4.2 Heterogeneous Effects by Student Characteristics

We examine heterogeneity in school closure effects along household income and prior academic performance (Section 4.2.1), and then summarize a second SES gradient by parental education. Heterogeneity by gender and parental education is reported in Appendix Tables B.11 and B.12.

4.2.1 Household Income and Academic Performance

Expenditure. Table 3 examines heterogeneity in expenditure by prior household income and academic performance, using terciles based on income and test score data at $g = -1$ for all students. Both dimensions show the presence of heterogeneous effects.

For household income (columns 2-4), low-income households raised their private tutoring expenditure by 6.2% per ten additional closure days, compared to just 2.9% and 2.8% for middle- and high-income households, respectively. Similarly, for prior academic performance (columns 5-7), low-performing students increased private tutoring investment by 7.0% per ten additional closure days, larger than the 3.4% for middle-performing students, a difference that is highly significant. The effect for high-performing students, 1.0%, is small and not statistically significant.

Table 3: Impact of School Closure Days on Expenditure: by Income and Performance

	Baseline	2019 Income			2019 Performance		
	(1) All	(2) Low	(3) Mid	(4) High	(5) Low	(6) Mid	(7) High
Closed School Days	0.041*** (0.005)	0.062*** (0.009)	0.029*** (0.008)	0.028*** (0.007)	0.070*** (0.010)	0.034*** (0.007)	0.010 (0.007)
p-value			0.007	0.901		0.005	0.014
N	17,104	5,886	6,338	4,880	5,702	5,702	5,700

Note: Clustered standard errors in parentheses (school level). P-values in columns (3)-(4) and (6)-(7) test whether effects differ from their adjacent left columns. For example, the p-value in column (3) tests the difference between columns (2) and (3), while the p-value in column (4) tests the difference between columns (3) and (4). ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

The larger *percentage* responses among lower-income and lower-performing households are partly a base effect: Appendix Table B.4 reports the same estimates in levels (10,000 KRW), and the absolute magnitudes are broadly similar across income groups (approximately 5,800 KRW for low, 3,650 KRW for middle, and 5,330 KRW for high), with no pairwise difference statistically significant. Thus, households substitute tutoring for

school days at similar absolute magnitudes regardless of income, while representing a larger budget share for lower-income households.²⁹

Table 4: Impact of School Closure Days on Test Score: by Income and Performance

	Baseline	2019 Income			2019 Performance		
	(1) All	(2) Low	(3) Mid	(4) High	(5) Low	(6) Mid	(7) High
Closed School Days	-0.006 (0.005)	-0.017*** (0.006)	0.004 (0.006)	-0.000 (0.006)	-0.006 (0.009)	-0.006 (0.005)	0.030*** (0.005)
p-value			0.010	0.603		0.966	0.000
N	17,104	5,886	6,338	4,880	5,702	5,702	5,700

Note: Clustered standard errors in parentheses (school level). P-values in columns (3)-(4) and (6)-(7) test whether effects differ from their adjacent left columns. For example, the p-value in column (3) tests the difference between columns (2) and (3), while the p-value in column (4) tests the difference between columns (3) and (4). ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Test scores. Table 4 reports the corresponding test-score heterogeneity. Students from low-income households are more adversely affected by school closures. For this group, ten additional closure days reduce test scores by 0.017 SD, while the effects for middle- and high-income students are small and statistically insignificant. This indicates that disadvantaged students rely more heavily on in-person public schooling. This “Great Equalizer” pattern is consistent with Hahn et al. (2023), who, using nationally representative Korean data, also find a null average effect alongside a widening of test-score inequality under school closures.

4.2.2 Parental education and Gender

Beyond income, students with less-educated parents are also disproportionately harmed by closures: ten additional closure days reduce test scores by 0.012 SD for students whose

²⁹We also show that all income terciles expand tutoring at comparable quantities and per-hour expenditures. This indicates that the larger *percentage* response among lower-income households reflects their lower pre-pandemic spending baseline rather than differential access or quality sorting. The corresponding level estimates and the income-specific quantity/quality decomposition are reported in Appendix Tables B.4, B.7, and B.8.

mothers have less than a four-year college degree and by 0.014 SD for students with below-BA fathers, while students with college-educated parents show null effects (Appendix Table B.11; differences across maternal-education groups). Expenditure responses, by contrast, are statistically indistinguishable across gender and parental-education subgroups (Appendix Table B.12). Aggregate economic disruption would itself have generated expenditure heterogeneity along these dimensions. Its absence is consistent with Fact 1 and supports the identification argument. We do not find significant heterogeneity by gender; these results are reported in Appendix Tables B.11 and B.12.³⁰

4.2.3 Joint Heterogeneity Analysis and Quantile Regression

Table 5 reports a fully interacted 3×3 subgroup analysis, splitting students by income and prior-attainment terciles. The negative test-score effects among low-income households are concentrated among middle-attainment students, while high-prior-attainment students experience positive test-score effects.

Table 5: Heterogeneous Effects by Income and Prior Attainment

Prior Attainment	Panel A: Expenditure (IHS)			Panel B: Expenditure (10K KRW)			Panel C: Test Scores (Std.)		
	Income Tercile			Income Tercile			Income Tercile		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
Low	0.097*** (0.016)	0.048*** (0.014)	0.047** (0.018)	1.013*** (0.314)	0.640* (0.331)	0.835 (0.519)	−0.008 (0.009)	0.009 (0.012)	−0.006 (0.013)
Mid	0.041*** (0.014)	0.039*** (0.012)	0.017 (0.012)	0.317 (0.253)	0.594 (0.395)	−0.130 (0.536)	−0.015** (0.007)	−0.003 (0.007)	0.002 (0.009)
High	0.022 (0.015)	−0.005 (0.011)	0.020** (0.009)	0.129 (0.317)	−0.147 (0.352)	0.713* (0.394)	0.027*** (0.006)	0.033*** (0.006)	0.030*** (0.007)

Note: Each cell reports the coefficient on closed school days from a separate regression estimated on the subsample defined by the intersection of baseline income tercile (columns) and baseline prior-attainment tercile (rows). In Panel A the dependent variable is the IHS of monthly tutoring expenditure; in Panel B it is monthly tutoring expenditure in 10,000 KRW; in Panel C it is the standardized test score. All specifications include student and year fixed effects and control for log income. Standard errors clustered at the school level in parentheses. Cell sample sizes (Low/Mid/High attainment within each income group): Low Income: 2,546, 1,958, 1,382; Mid Income: 1,968, 2,216, 2,154; High Income: 1,188, 1,528, 2,164. */**/** indicate significance at the 10/5/1 percent level.

³⁰This absence of gender bias in the educational investment response is consistent with the empirical findings in the literature, given that Korea's historical son preference has declined over the last two decades (Chung and Gupta 2007; Yoo et al. 2017; Choi and Hwang 2020).

Additionally, Appendix Table B.15 reports conditional quantile regressions as supplementary distributional evidence. Because these specifications cannot include student fixed effects in our two-period panel, we treat them as descriptive rather than as our primary estimates. The results support the school-level dispersion findings: expenditure responses are not concentrated in a particular part of the spending distribution once measured in levels, while test-score gains are concentrated outside the lower tail.

4.3 School-Level Effects

In this section, we turn to distributional outcomes using Equation (2). Given that our treatment varies at the school level, we focus on the within-school standard deviation of test scores to assess whether school closures widened educational inequality. Table 6 presents how school closures affected educational inequality by using within-school standard deviations of expenditure and test scores.³¹ For time-varying controls, we use school-level averages of parental expenditure and household income after IHS transformation. As with our individual-level results, our preferred specifications include time-varying controls (columns 2 and 5). Appendix Tables B.13 and B.14 report how these school-level estimates change with different specifications.

School closures significantly affected test score dispersion: ten additional school closure days increased the within-school standard deviation of test scores by approximately 0.015 standard deviations, which is approximately a 25% increase. This effect is statistically significant at the 5% level with controls, robust against controlling for average tutoring expenditure (column 5), and the interaction term with the above-median closure days indicator is virtually zero (column 6). That is, even if average test scores remained unaffected during school closures (Section 4.1), the distribution of achievement significantly widened. The corresponding test-score quantile regressions (Appendix Table B.15, Panel B) provide detailed additional evidence. Closures have a near-zero effect at the

³¹We add the city fixed effects on top of school and grade fixed effects.

Table 6: Impact of School Closure Days on School-level Standard Deviation

	Expenditure			Test Score		
	(1)	(2)	(3)	(4)	(5)	(6)
Closed School Days	-0.249 (0.246)	-0.249 (0.247)	-0.269 (0.258)	0.014* (0.007)	0.015** (0.007)	0.015** (0.007)
IHS(income)		0.683 (7.655)	0.598 (7.675)			
IHS(expenditure)					-0.001* (0.001)	-0.001* (0.001)
Above Median=1 × Closed School Days			0.059 (0.240)			0.000 (0.004)
<i>N</i>	1,206	1,206	1,206	1,206	1,206	1,206

Note: The dependent variables are within-school standard deviations of tutoring expenditure and test scores. Controls are school-level averages of IHS household income and IHS tutoring expenditure, as indicated. Standard errors clustered at the city level in parentheses. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

10th percentile of the conditional achievement distribution but raise scores by +0.027 to +0.044 SD at Q25–Q90, so the lower tail is unaffected while the upper tail gains. This is consistent with a widening of the within-school SD. On the other hand, the closures had little impact on the standard deviation of parental expenditure within schools. The effect is insignificant and remains unaffected when controlling for average income and the above-median closure indicator.

A remaining concern is that closure intensity may be correlated with school quality or other school-level characteristics. This does not appear to be the case: school closure days are not correlated with school quality, proxied by per-capita spending (Figure 2). In addition, our individual-level specifications include student fixed effects, which absorb time-invariant differences in student background and baseline school composition. This is consistent with South Korea's school equalization policies, under which students are assigned to middle schools and most high schools by lottery conditional on residence.³²

³²This feature has been exploited by various papers in the literature. See, for example, Kang (2007), Park et al. (2013), and Park et al. (2018)

Students are assigned to middle schools (our 8th graders) and high schools (our 11th and 12th graders) via lottery, conditional on residence.³³

4.4 Implications for the Structural Model

Our analysis points to three key findings. First, when schools closed, households increased private tutoring as a substitute, with similar absolute magnitudes across income groups, which represented a larger budget share for lower-income households. Second, despite this compensatory response, the test score distribution widened. School closures directly harmed lower-income students, who rely heavily on public education, while higher-attaining students' scores actually increased as they substituted into more effective private tutoring, indicating that private tutoring cannot replicate the equalizing role of public schooling. Third, school closures increased the dispersion of test scores within school.

These findings indicate that the relevant heterogeneity for inequality operates through two distinct channels: the household's financial capacity to substitute (income-driven expenditure) and the direct return to in-person instruction (prior-attainment-driven test scores). Separating and quantifying each requires a structural framework. This leads to several questions: how does each channel contribute to the changes in the unconditional dispersion of test scores? And, when tutoring support is offered, how large a targeted voucher is sufficient to reduce inequality? We develop a structural model to address these questions in the following section.

5. Theoretical Framework

Motivated by the empirical results, we employ the tournament model of Kang (2024) in which each household makes investments in their child. We extend this framework to

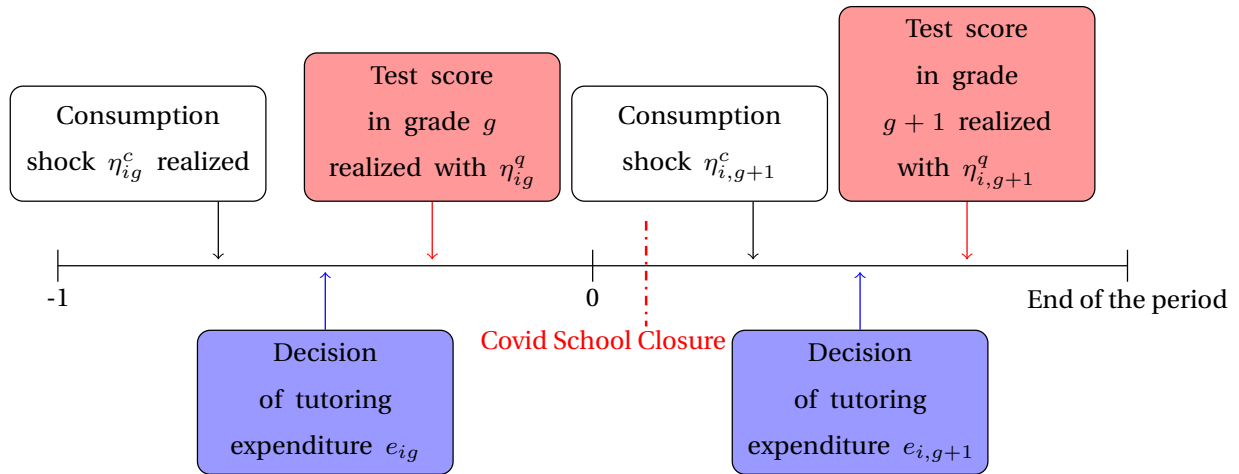
³³Specialized high schools can select students, but they account for just 3% of total enrollment.

capture three key elements: (1) the underlying incentives driving demand for private tutoring, (2) the effects of private tutoring and school attendance on test scores, and (3) the substitution relationship between these educational inputs. The tournament structure represents the competitive college admissions process, where parental investments are driven by the potential returns to educational achievement through access to selective institutions. Test score production follows a Constant Elasticity of Substitution (CES) function incorporating three inputs: the child's previous academic performance, in-person school days, and private tutoring expenditures. This specification enables us to identify both the marginal effects of each educational input and the elasticity of substitution between public schooling and private educational investments when schooling availability changes exogenously.

5.1 Environment

Consider an economy composed of N households with a unitary decision making process. Figure 4 presents our model timeline. The model begins in period -1, the year preceding the school-closure year (for treated groups only). The initial conditions are household income and the test score from the previous grade.

Figure 4: Model Timeline



Denote g as the school grade of the student. Each household i derives flow utility from current consumption and the perceived returns to academic outcomes, R_{ig} , which is a function of the test score $q_{i,g+1}$ and the reward determined following tournament structure (Kang 2024). The sequence within each period is as follows. First, a consumption shock η_{ig}^c is realized. Based on this shock and expectations about the test score shock η_{ig}^q , each household makes a private tutoring investment decision e_{ig} . After this investment decision, the test score $q_{i,g+1}$ is produced with the realization of the time-specific test score shock η_{ig}^q . The generated test score $q_{i,g+1}$ then serves as an input for producing the subsequent test score in the next period.

5.2 Household and the Tournament Structure

Each period, household i derives utility from consumption and the test score produced in grade g . The utility function is defined as:

$$\eta_{ig}^c \ln(c_{ig}) + \alpha_g R(q_{i,g+1}), \quad (4)$$

where c_{ig} is household consumption, $R(q_{i,g+1})$ is the return from test score $q_{i,g+1}$, η_{ig}^c is a shock to the marginal utility of consumption, and α_g is the grade-specific preference weight on the child's expected educational return.

We adopt a static tournament model to capture the underlying demand for parental investment.³⁴ Each household decides how much to spend on private tutoring expenditure e_{ig} to maximize utility defined in equation (4), subject to the budget constraint:

$$c_{ig} + e_{ig} \leq w_{ig} \quad (5)$$

³⁴An alternative absolute-performance model overpredicts parental investment. See Figure C.1.

where w_{ig} is household income. The tournament structure is realized through the return function $R(q_{i,g+1})$, which is specified as:

$$R(q_{i,g+1}) = \sum_{j=1}^J \ln(v_j) \times \text{Prob}(\ln \tilde{Q}_{j-1} \geq \ln q_{i,g+1} \geq \ln \tilde{Q}_j \mid \Gamma_{ig}) \quad (6)$$

where $\tilde{Q}_j$ is the cutoff for college tier j , v_j is the lifetime income associated with graduating from college tier j , and Γ_{ig} is the information set of household i when their child is in grade g . Competition between households occurs around the cutoff $\tilde{Q}_j$. Let m_j denote the number of seats in college tier j , and let $M_j = \sum_{k=1}^j m_k$ denote the cumulative number of seats through tier j . The cutoff $\tilde{Q}_j$ is the test score of the student ranked M_j . Thus, students with scores between $\tilde{Q}_{j-1}$ and $\tilde{Q}_j$ are assigned to college tier j .

This return function assumes that households value the probability that their child reaches a higher college tier. These probabilities differ across students because prior achievement, household resources, school days, and tutoring choices shift the distribution of future test scores. Students with stronger predicted achievement are more likely to cross higher-tier cutoffs, while students with weaker predicted achievement are more likely to fall into lower tiers.

5.3 Test Score Production Function

To map parental investment and school days to test scores, we specify a production function:

$$q_{i,g+1} = f(\theta_g, q_{ig}, e_{ig}, schdays_{ig}, \eta_{ig}^q) \quad (7)$$

where θ_g is a set of relevant parameters that may change over school grade (age). To identify substitution between parental investment and school days, our functional form follows the constant elasticity of substitution (CES) production function with constant

returns to scale (Cunha et al. 2010):

$$q_{i,g+1} = A_g \left[\delta_{q,g} q_{ig}^\phi + \delta_{s,g} schdays s_{ig}^\phi + \delta_{eg} (1 + e_{ig})^\phi \right]^{\frac{1}{\phi}} \eta_{ig}^q, \quad (8)$$

where A_g is the total factor productivity, ϕ is the substitution parameter, $\delta_{q,g}$ is the marginal effect of the previous test score, $\delta_{s,g}$ is the marginal effect of school days, δ_{eg} is the marginal effect of private tutoring expenditure, and η_{ig}^q is the time-varying shock to the test score.

5.4 Unobserved Heterogeneity

We allow for persistent unobserved heterogeneity across households using a discrete mixture of $K = 4$ latent types, following Heckman and Singer (1984). Each household i is assigned a fixed type $k_i \in \{1, 2, 3, 4\}$ with probabilities $(\pi_1, \pi_2, \pi_3, \pi_4)$, where $\sum_{k=1}^4 \pi_k = 1$.

Types shift the location of the consumption shock and test-score productivity. For type k ,

$$\log \eta_{ig}^c = \lambda_c^k + \tilde{\eta}_{ig}^c, \quad \tilde{\eta}_{ig}^c \sim \mathcal{N}(0, \sigma_c^2),$$

and

$$q_{i,g+1}^k = \exp(\lambda_q^k) A_g \left[\delta_{q,g} q_{ig}^\phi + \delta_{s,g} schdays s_{ig}^\phi + \delta_{eg} (1 + e_{ig})^\phi \right]^{1/\phi} \eta_{ig}^q.$$

The four types are the 2×2 combinations of two scalar shifters: $(0, 0)$, $(\bar{\lambda}_c, 0)$, $(0, \bar{\lambda}_q)$, and $(\bar{\lambda}_c, \bar{\lambda}_q)$, with type 1 as the reference type.

5.5 Model Solution

Each household chooses tutoring expenditure e_{ig} to maximize (4) subject to the budget constraint (5) and the production function (8). Substituting the budget constraint and

production function, the first-order condition with respect to e_{ig} is

$$\underbrace{-\frac{\eta_{ig}^c}{w_{ig} - e_{ig}}}_{\text{marginal cost}} + \underbrace{\alpha_g \varphi_{ig} \frac{\delta_{eg}(1 + e_{ig})^{\phi-1}}{B_{ig}}}_{\text{marginal benefit}} = 0, \quad (9)$$

where $B_{ig} = \delta_{q,g} q_{ig}^\phi + \delta_{s,g} schdays_{ig}^\phi + \delta_{eg}(1 + e_{ig})^\phi$ is the CES aggregate inside the production function, and $\varphi_{ig} = \sum_{j=1}^{J-1} \left(\ln v_j - \ln v_{j+1} \right) \frac{1}{\sigma_q} \phi \left(\frac{\ln \tilde{Q}_j - \ln \widehat{q_{i,g+1}}}{\sigma_q} \right)$ is the marginal return to test scores through the tournament structure, which weights the prize difference between adjacent tiers $(\ln v_j - \ln v_{j+1})$ by the density of the student's predicted test score near each cutoff $\tilde{Q}_j$. The first term in (9) captures the marginal utility cost of diverting income from consumption to tutoring. The second term captures the marginal benefit: spending on tutoring raises the predicted test score through the production function, which in turn increases the expected tournament return. In the interior solution ($e_{ig} > 0$), the household equates these two margins. When the marginal cost exceeds the marginal benefit at $e_{ig} = 0$, the household does not participate in tutoring. Appendix D provides the full derivation.

6. Structural Estimation

In this section, we first explain the penalized maximum-likelihood estimation strategy. We then discuss the identification of the structural parameters, present the structural estimates and model fit, and examine sensitivity to the penalty weights.

6.1 Maximum Likelihood

Individual likelihood contribution The individual likelihood contribution is composed of the contributions regarding private tutoring expenditure e_{ig} and test scores q_{ig} for $g = -1, 0$. Denoting θ as the set of parameters, each household i 's likelihood contribution

is

$$\mathcal{L}_i(\theta|q_{i,-1}, w_{i,-1}, w_{i0}; k) = \prod_{g=-1}^{g=0} \mathcal{L}_{ig}(\theta|q_{ig}, w_{ig}; k), \quad (10)$$

and

$$\begin{aligned} \mathcal{L}_{ig}(\theta|q_{ig}, w_{ig}; k) &= \left[f_{e_{ig}}(e_{ig}|k) \cdot f_{q_{i,g+1}}(q_{i,g+1}|e_{ig}, k) \right]^{d_{ig}^e} \\ &\times \left[\Pr(e_{ig} = 0|k) \cdot f_{q_{i,g+1}}(q_{i,g+1}|e_{ig}, k) \right]^{(1-d_{ig}^e)} \end{aligned}$$

where d_{ig}^e is an indicator function of tutoring participation decision (i.e. $e_{ig} > 0$ if $d_{ig} = 1$), and conditioning on the latent type k enters through the type-specific shifters $(\lambda_c^k, \lambda_q^k)$. Since the type is unobserved, the household-level contribution is obtained by marginalizing over the discrete type distribution:

$$\mathcal{L}_i(\theta|q_{i,-1}, w_{i,-1}, w_{i0}) = \sum_{k=1}^4 \pi_k \mathcal{L}_i(\theta|q_{i,-1}, w_{i,-1}, w_{i0}; k).$$

The unrestricted part of the likelihood function is

$$\mathcal{L}(\theta) = \prod_{i=1}^N \mathcal{L}_i(\theta|q_{i,-1}, w_{i,-1}, w_{i0}).$$

Appendix D provides technical details on evaluating the likelihood contributions using the corresponding shocks and the Jacobian transformation.

Aggregate restrictions through a penalty We augment the likelihood with aggregate restrictions that mirror the empirical design. This follows earlier structural work that disciplines estimation using auxiliary moments or aggregate restrictions: Sieg and Yoon (2022) augment the likelihood with penalty functions so predicted distributions match non-parametric estimates, and Kang (2016) impose aggregate moment conditions to

discipline structural parameters. In our setting, the restrictions require simulated data to reproduce the reduced-form TWFE estimates and to satisfy the parallel-trends restriction underlying the difference-in-differences design.

Specifically, we add two penalties to the log likelihood:

$$\hat{\theta} = \arg \max_{\theta} \left[\log \mathcal{L}(\theta) - \lambda_{\text{TWFE}} \mathcal{C}_{\text{TWFE}}(\theta) - \lambda_{\text{parallel}} \mathcal{C}_{\text{parallel}}(\theta) \right].$$

Here, $\mathcal{C}_{\text{TWFE}}(\theta)$ measures the distance between the model-simulated TWFE coefficients and their reduced-form counterparts for tutoring expenditure, test scores, and within-school dispersion. The term $\mathcal{C}_{\text{parallel}}(\theta)$ measures the distance between the model-implied parallel-trend conditions and the corresponding reduced-form estimates from the control cohorts. The weights λ_{TWFE} and $\lambda_{\text{parallel}}$ govern how tightly each set of restrictions is enforced. We examine sensitivity to these weights in Section 6.4.

6.2 Identification

We specify parametric forms for both the production function and the flow utility function, allowing some parameters to vary across cohorts and grades and accommodating discrete unobserved heterogeneity across households, while imposing restrictions on the remaining parameters to ensure model identification.

For the household preference, α_g is identified by the average level of private tutoring expenditure, which is allowed to differ by cohorts. The variance σ_c^2 of the time-varying consumption shock η_{ig}^c is identified by the within-household residual variation in income net of tutoring conditional on the household's unobserved type.

In the production function, we assume that the substitution parameter ϕ , the relative effect of school days κ , and the variance σ_q^2 of the time-varying test score shock η_{ig}^q are constant across the population. The covariation between average tutoring expenditure and the number of school days, and the covariation of these inputs with subsequent test

scores jointly identify ϕ . The relative weight of school days compared to private tutoring κ is identified by the covariation of subsequent test scores and the number of school days. The variance σ_q^2 is identified by the residual variation not explained by the observable inputs in the CES production function in equation (7).

To capture time-varying heterogeneity across different grade levels and cohorts, we allow several production-function parameters to vary by grade g and cohort c . Specifically, we allow the constants (A), the CES weight of tutoring (δ_e), and the CES weight of school days (δ_s) to vary by cohorts and grades. These parameters are identified by covariation between the inputs and subsequent test scores, for which we have year-cohort variations. We however maintain that the ratio of these parameters remains constant over time, so that $\delta_{sg}^c / \delta_{eg}^c = \kappa$. This restriction stems from the limited variation available for cohorts who did not experience the Covid-19 school closure.

Finally, the discrete type probabilities ($\pi_1, \pi_2, \pi_3, \pi_4$) and the persistent shifters ($\bar{\lambda}_c, \bar{\lambda}_q$) are identified by the repeated pattern of each household across two periods. This separates the persistent type-specific components from the i.i.d. time-varying shocks. Households whose consumption patterns (income net of tutoring) persistently deviate in the same direction across periods contribute to the identification of $\bar{\lambda}_c$; households whose test scores persistently exceed or fall short of the production function's prediction contribute to $\bar{\lambda}_q$. The mixing probabilities are identified by the relative frequencies of these persistent deviation patterns in the data.

6.3 Structural Estimates and Model Fit

We present the structural estimates and explain what they imply for substitution between school days and tutoring expenditure and for differences across cohorts in the effect of tutoring expenditure on test scores. We then examine the model fit and analyze the sensitivity of the results to the moment penalty.

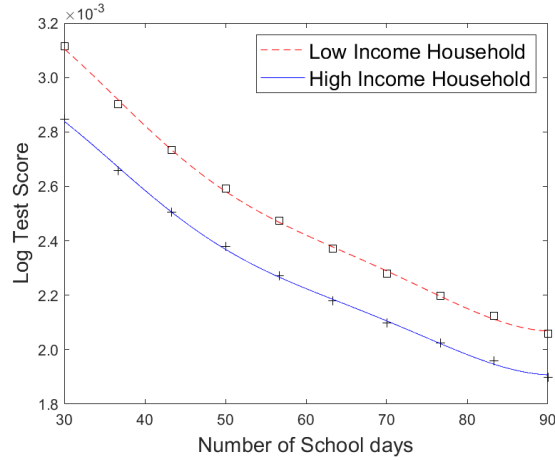
Table 7: Structural Parameter Estimates

Parameter	Description	Estimate	S.E.
σ_q	Test score error	0.121	(0.000)
σ_c	Consumption error	1.520	(0.003)
κ	Home production ratio	0.419	(0.001)
ϕ	CES substitution	0.721	(0.001)
Unobserved Heterogeneity (Heckman–Singer)			
$\bar{\lambda}_c$	Consumption shifter	0.863	(0.005)
$\bar{\lambda}_q$	Test score production shifter	-0.284	(0.001)
π_1	Type 1 probability (0, 0)	0.435	(0.001)
π_2	Type 2 probability ($\bar{\lambda}_c$, 0)	0.270	(0.001)
π_3	Type 3 probability (0, $\bar{\lambda}_q$)	0.254	(0.001)
π_4	Type 4 probability ($\bar{\lambda}_c$, $\bar{\lambda}_q$)	0.040	—

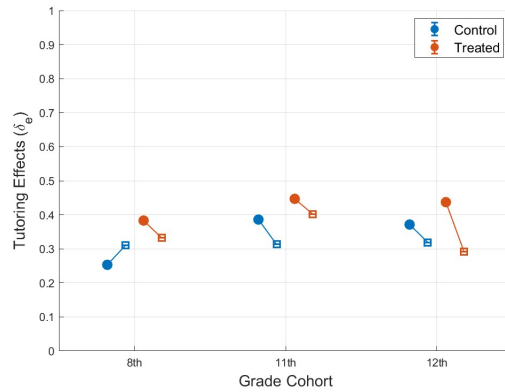
Note: Presented parameters are common across cohorts and grades. Standard errors in parentheses computed using delta method.

Table 7 reports the estimates of time-invariant structural parameters. The CES substitution parameter ϕ is 0.721, implying that school days and tutoring are substitutable inputs in the production of test scores. This substitutability is key to understanding the household response. Since tutoring can partially compensate for lost school days, households facing school closures optimally increase tutoring expenditure, consistent with our reduced-form evidence. However, similar absolute increases in tutoring need not imply equal compensation: low-income students may rely more heavily on in-person schooling, so the same tutoring response may offset less of the achievement loss. Consistent with this mechanism, Figure 5 shows that the marginal product of school days is higher for low-income households. This aligns with our reduced-form finding that low-income households experience larger achievement losses despite actively increasing tutoring investment.

The estimate of the relative CES weight of school days to tutoring expenditure, $\hat{\kappa}$, is 0.417 (Table 7). Figure 6 shows that the marginal productivity of tutoring expenditure, $\hat{\delta}_e$,

Figure 5: Marginal Effects of School Days

Note: Marginal effects are the first-order derivative of log test score with respect to schdays. Effects are depicted separately for high-income households (above the 80th income percentile) and low-income households (below the 20th income percentile).

Figure 6: Tutoring Effects by Grade Cohort and Treatment Group

Note: This figure illustrates the estimated tutoring effects (δ_e) across three grade cohorts (8th, 11th, and 12th grades). Estimates for control and treated groups at two periods (g and $g + 1$) are presented side by side, with distinct markers indicating groups (circles for control period g , squares for control period $g + 1$, diamonds for treated period g , triangles for treated period $g + 1$). Error bars represent standard errors of the estimates.

lies between 0.25 and 0.45 in every cohort–grade cell. Multiplying these $\hat{\delta}_e$ values by $\hat{\kappa}$ implies an associated marginal effect of a school day, $\hat{\delta}_s$, of about 0.10 to 0.19. In words, we do not observe systematically higher marginal effects of tutoring for the cohorts that faced COVID-19 school closures (“treated”) compared to others (“control”), which suggests that the tutoring response to school closures is driven by input substitution under binding budgets rather than cohort heterogeneity in the tutoring effect. The full set of structural estimates is reported in Table D.1.

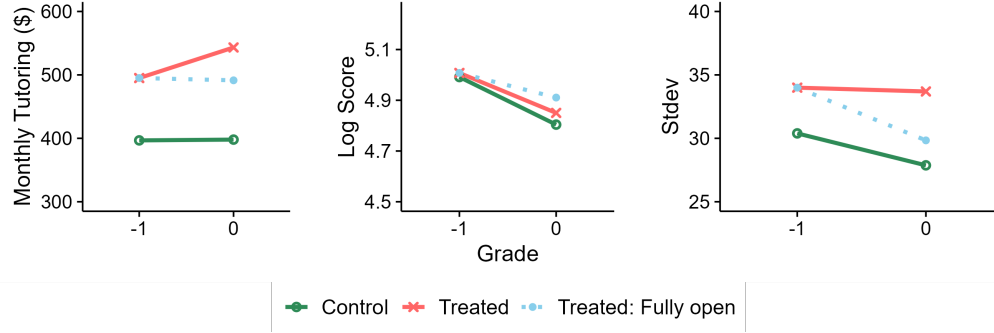
Table 8: TWFE Moments Fit

	Data	Model
Coefficient on schdays in log tutoring	-0.0410	-0.0464
Coefficient on schdays in log test score	0.0016	0.0003
Coefficient on schdays in stdev	-0.2700	-0.2757

Note: This table presents the fit of the TWFE regression coefficients for data and simulation. The corresponding reduced form estimates are Tables 2, 6, and B.10.

Finally, we examine the model fit and the implementation of the parallel trends assumption. First, the model fits the TWFE estimates. Table 8 presents the fit of the TWFE estimates of the previous section and the simulated TWFE within the model, which shows an excellent fit. The model also captures the key data pattern for different cohort and treated control groups, which is reported in Table D.2. Secondly, imposing parallel trend clearly demonstrates the difference between the treated vs control groups. Figure 7 presents simulated tutoring expenditure under the parallel-trends constraint. For the cohort exposed to COVID-19 school closures, tutoring expenditure increases more steeply, and the dispersion of test scores increases. The dotted lines represent the counterfactual scenario if the schools had remained fully open. Tutoring expenditure would have decreased significantly, following the trends of the control group, and school-level inequality would not have risen as substantially in the absence of the school closures.³⁵

³⁵We discuss this as a counterfactual scenario in Section 7.

Figure 7: Trends of Simulated Outcome Variables

Note: This figure presents the implementation of a parallel trend in the structural estimation results using simulated monthly tutoring expenditure. Specifically, it shows the trends of average tutoring expenditure for three different cohorts as students progress from g to $g + 1$.

6.4 Sensitivity Analyses

We examine how the structural estimates vary with the choice of penalty weights. We re-estimate the model across a 3×3 grid of $\lambda_{\text{TWFE}} \in \{0, 500, 1500\}$ and $\lambda_{\text{parallel}} \in \{0, 250, 500\}$, holding the parallel-trend-in-school-dispersion weight fixed at $\lambda_{\text{par,sd}} = 50,000$. The structural parameters of interest remain similar across the grid: $\hat{\phi}$ ranges from 0.70 to 0.74 (baseline 0.721), $\hat{\kappa}$ from 0.41 to 0.44 (baseline 0.419), $\hat{\sigma}_c$ from 1.49 to 1.55, $\hat{\sigma}_q$ is essentially constant at 0.12, and the type shifters remain in narrow ranges, with $\bar{\lambda}_c \in [0.86, 0.97]$ and $\bar{\lambda}_q \in [-0.29, -0.28]$.

The penalty terms mainly discipline the model-implied reduced-form moments. In the unrestricted cell, the model predicts a positive tutoring response, but its magnitude is below the reduced-form estimate (0.006 versus 0.041). Under the baseline weights $(\lambda_{\text{TWFE}}, \lambda_{\text{parallel}}) = (1500, 500)$, the model closely matches the targeted moments: the simulated coefficients are 0.046 for tutoring, 0.000 for test scores, and 0.277 for within-school dispersion, compared with data targets of 0.041, -0.002 , and 0.270.

7. Policy Experiments

Tutoring vouchers are increasingly used as a tool to help disadvantaged students recover from learning disruptions. England’s National Tutoring Programme and South Korea’s 2020 Future Education Center budget exemplify real-world initiatives expanding access to tutoring. Using the estimated model, we first simulate a counterfactual full-open scenario in order to quantify the impact of school closures on students’ educational outcomes, and use it as the benchmark for evaluating the voucher policy. We then conduct ex-ante evaluations of voucher programs and assess their distributional consequences.

7.1 School Reopening and Education Inequalities

Table 9: Baseline vs. Full Open (All Individuals)

	Baseline	Full Open
Mean Tutoring Expenditure	63.95	57.85
Mean Log Test Score	4.8503	4.9107
SD Log Test Score	0.3170	0.2919

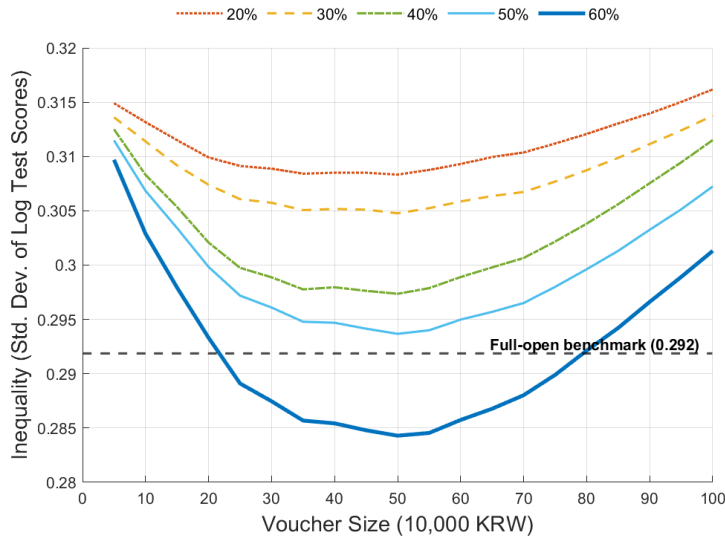
Note: This table presents a comparison between the baseline simulation (Baseline) and the counterfactual simulation where in-person school days total 95 days (Full Open).

We first simulate a full-open counterfactual by increasing school days to the pre-pandemic level of 95 days. Table 9 shows that, relative to the baseline with actual closures, the full-open counterfactual reduces the standard deviation of log test scores by 7.9% and increases the mean log score by 1.2%. The larger effect on dispersion than on the mean mirrors our reduced-form findings that school closures reduce test scores for low-income students while leaving middle- and high-income students largely unaffected (Section 4), and is consistent with the higher marginal product of school days for low-income households shown in Figure 5.

7.2 Targeted Tutoring Voucher

Motivated by the unequal effects of school closures documented both in TWFE and the structural estimates, we evaluate whether targeted tutoring vouchers can compensate for the inequality generated by school closures. Figure 8 presents the relationship between voucher size and test-score inequality across eligibility thresholds. The 60th income percentile is the narrowest eligibility threshold under which full recovery is achievable.³⁶

Figure 8: Tutoring Voucher and Test Score Inequality



Note: Each curve represents a different eligibility threshold (percentage of households eligible for the voucher by income). The horizontal dashed line represents inequality under full school reopening (standard deviation of 0.292).

Two findings emerge. First, a voucher of approximately 250,000 KRW targeted at the bottom 60% of the income distribution is sufficient to fully offset the inequality increase caused by school closures: at this amount, test-score dispersion returns to the level that would prevail under full school reopening (0.292). More narrowly targeted policies (20–50% eligibility) cannot reach this benchmark regardless of voucher size. The 60% eli-

³⁶The Korean Ministry of Education implemented the Future Education Center tutoring voucher program, which subsidized 500K KRW (around \$400 USD) per student through group tutoring. Tutors were primarily college students, with most groups matching four students to each tutor.

gibility threshold is consistent with Table 5, where the joint heterogeneity analysis shows especially large achievement losses among low-income middle-performing students.

Second, the relationship between voucher size and inequality is non-monotonic. Beyond approximately 500K KRW, additional voucher spending actually *increases* inequality. This occurs because excessively large subsidies concentrated among eligible households raise their test scores disproportionately, which widens the gap between recipients and non-recipients. The inverted-U pattern is present at all eligibility thresholds and highlights that more generous vouchers are not necessarily more equalizing.

We indirectly assess the external validity of the voucher policy by varying the steepness of college-tier returns. We recompute the counterfactual under perturbations of the prize vector $\log v_j$ by a multiplier $\alpha \in [0.25, 1.50]$, where $\alpha = 1$ is the Korean baseline. For modest perturbations around Korea ($\alpha \in \{0.75, 1.00, 1.25\}$), the 250K KRW voucher target is unchanged. Under a substantial flattening ($\alpha = 0.25$, comparable to weak-elite-premium settings such as much of Europe), the closure-induced inequality damage shrinks to roughly a third of Korea's. In that case, a smaller 150K KRW voucher restores the modest remaining gap (Appendix Figure B.2). The mechanism we identify could be extended to other educational environments. Whether the voucher policy is worth pursuing, however, depends on whether the local college-tier returns are steep enough to generate meaningful inequality effects.

Crowd-out of private tutoring expenditure. A related policy concern with publicly funded tutoring is whether the subsidy crowds out private investment that households would have made on their own. Appendix Figure B.1 shows that crowd-out is limited at the policy-relevant voucher size: under 60% eligibility, a 250,000 KRW voucher leaves average private tutoring expenditure among eligible treated households essentially unchanged. Larger vouchers, however, do crowd out private spending. A 500,000 KRW voucher reduces private expenditure by about 102,000 KRW, and a 1 million KRW voucher

reduces it by about 290,000 KRW. Thus, crowd-out is modest at the voucher size that restores test-score inequality, but becomes meaningful for more generous subsidies. This pattern is consistent with the empirical literature on public–private investment relationships in education, which finds mixed evidence and, in many settings, imperfect substitutability between public and parental investments (Becker and Tomes 1986; Todd and Wolpin 2003; Datar and Mason 2008; Das et al. 2013; Fredriksson et al. 2013; Greaves et al. 2023).

8. Conclusion

This paper investigates how school closures affect educational outcomes through parental responses. We estimate the causal impact of school closures using quasi-experimental variation in school closure days in South Korea during the COVID-19 pandemic. Building on these empirical findings, we develop and estimate a tournament model to conduct policy counterfactual experiments. Our analysis highlights the equalizing role of public education and suggests that targeted voucher policies can be an effective alternative when full school reopening is not feasible. These findings show the need to account for heterogeneity when designing compensatory education policies.

Our empirical design allows us to identify the causal effect of lost in-person schooling on parental investment and educational inequality. These reduced-form findings therefore speak beyond the Korean context to the broader question of how households respond when public schooling is disrupted. The quantitative implications of the structural counterfactuals, however, are most directly informative for settings with similar institutional features. In countries with weaker tutoring infrastructure, flatter returns to elite higher education, or more discretionary attendance, the effectiveness of tutoring vouchers may differ.

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A. Data Appendix

A.1 Descriptive Statistics

Table A.1: Descriptive Statistics

	8th graders		11th graders		12th graders	
	Treated	Control	Treated	Control	Treated	Control
N (Students)	3,572	3,045	2,063	1,353	2,740	1,760
N (Schools)	121	56	102	41	379	63
Female (%)	46.9	45.9	47.5	53.5	50.5	51.5
Mother's Education (%)						
2-year College or Less	61.51	68.03	65.30	55.70	61.51	68.03
4-year College or More	38.49	31.97	34.70	44.30	38.49	31.97
Average HH Income	4,657	4,516	5,141	4,880	5,109	4,351
Private Tutoring Expenditure						
Korean	42.9	45.9	97.8	112.2	133.0	106.6
Mathematics	175.6	138.9	207.4	205.8	216.3	193.8
English	172.4	140.7	250.2	241.0	264.4	232.4

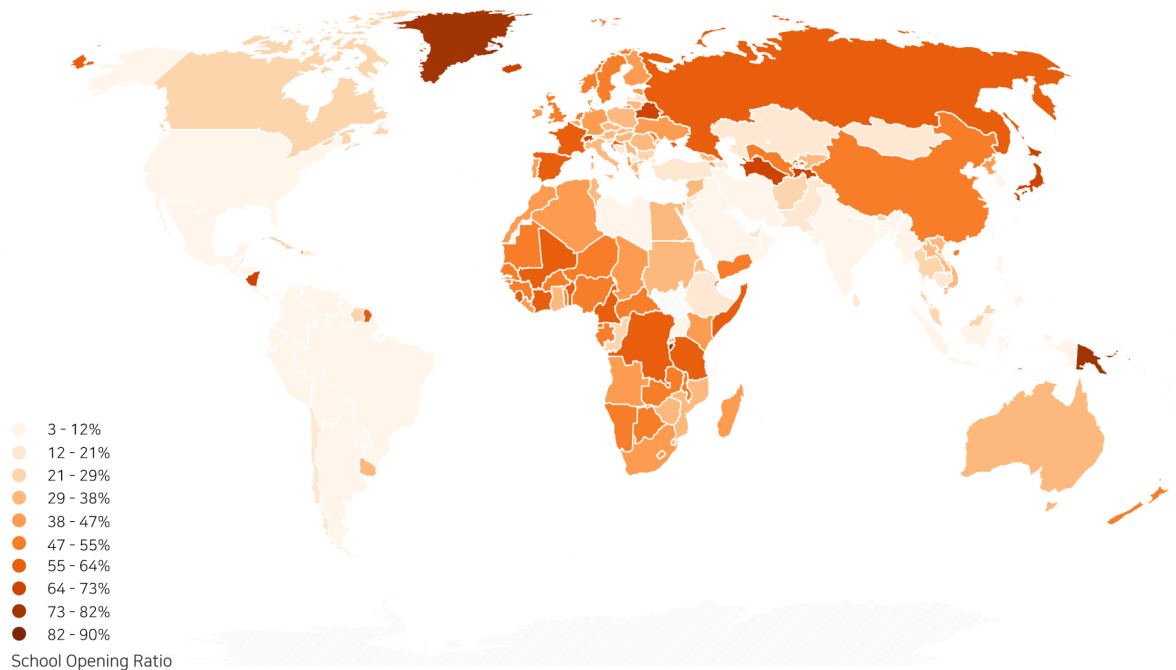
Note: Standard deviations in parentheses. All USD values are converted from KRW. (1 USD = 1,200 KRW) HH income and private tutoring expenditure are monthly values. “Selective school” is a sum of “autonomous” and “special purpose” schools. All numbers are calculated from the final year of each cohort.

A.2 Additional Data Description

Each year, for both surveys, information was collected through four separate questionnaires: one for teenagers, one for parents, one for teachers, and one for school principals. The survey data is then merged with administrative school information, including the number of teachers, students, classrooms, and campus size, among other details. This comprehensive design covers household information such as parental education, occupation, and income, providing valuable insights into the relationships between household composition and the educational factors of parents, including their education, occupation, and income.

The datasets contain detailed information on students' learning behaviors. In particular, they ask detailed information about participation in private tutoring, their types (e.g., group meetings, one-on-one, online), their topics (e.g., Korean, English, math), the hours spent on them and their cost. Importantly, the datasets offer information about academic achievement of adolescents, measured by dedicated cognitive tests (Middle School Sample) and national-level achievement test (High School Sample). It enables us to map the student decisions to the changes in academic achievement.

Figure A.1: Global School Opening Ratio During the Pandemic (2020-2021)



Data: World Bank

Figure A.2: Spatial variations in School Closure Days

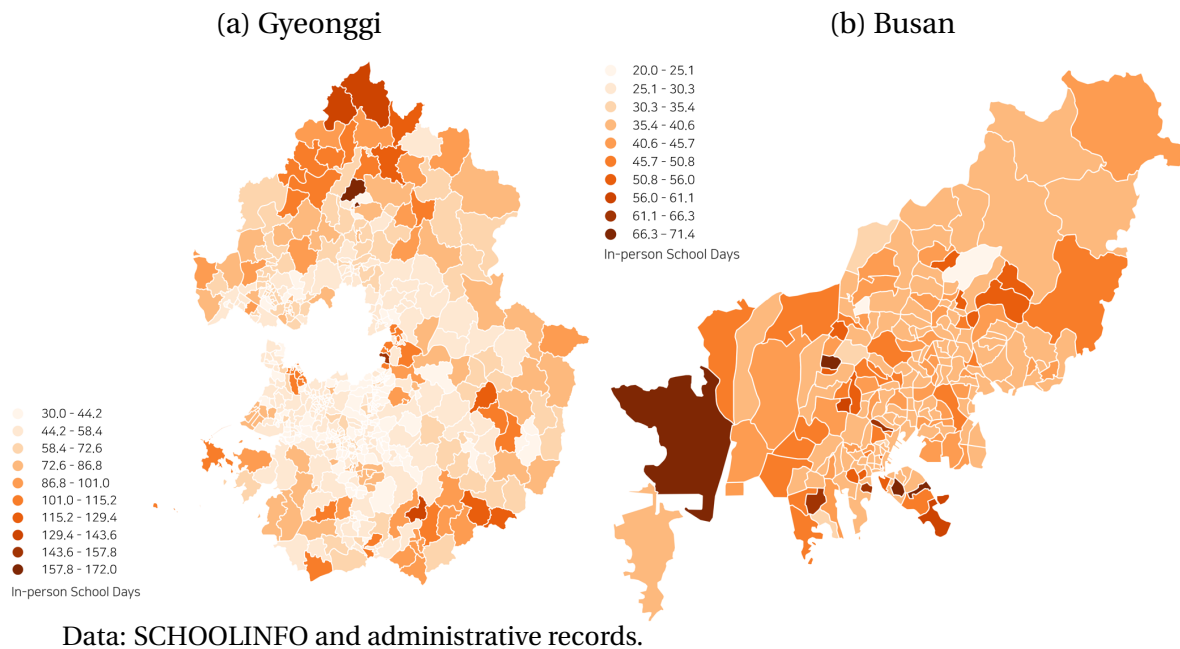


Figure A.3: Monthly Employment by Country, 2020

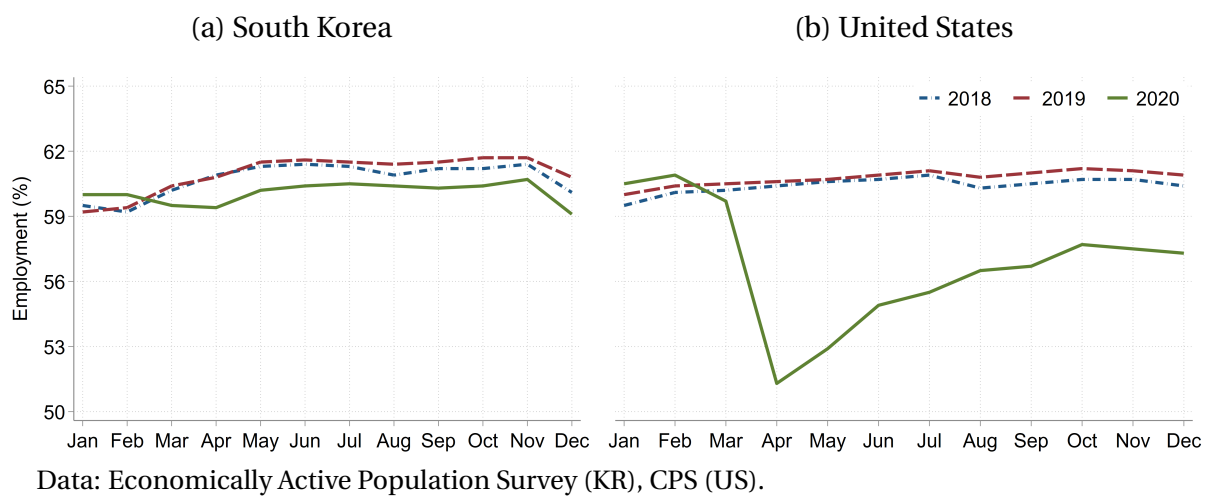
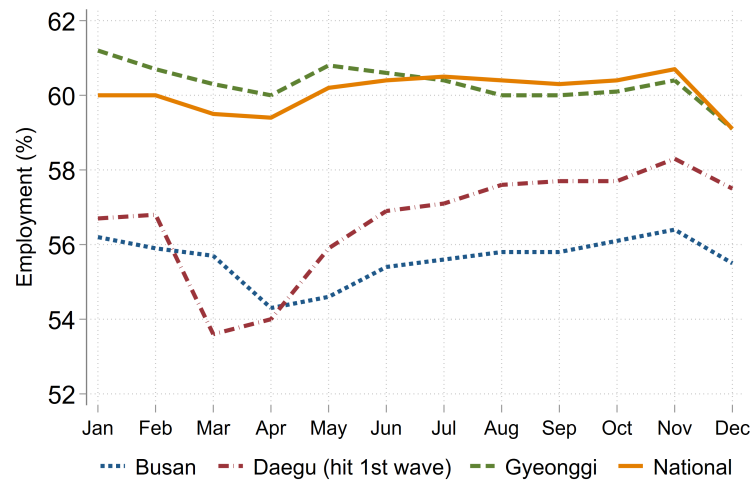
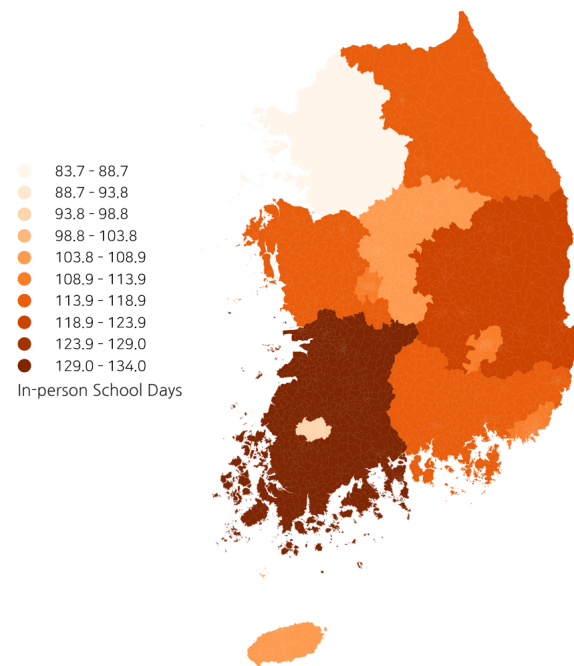


Figure A.4: 2020 Monthly Employment of South Korea, by Province



Data: EAPS.

Figure A.5: School Closure Days in South Korea (Province Level)



Data: SCHOOLINFO and administrative records.

B. Additional Empirical Results

Table B.1: IV-Style Diagnostic Using Local COVID Incidence as Instrument

Dep. variable:	IV/2SLS		First Stage		OLS	
	(1)	(2)	(3)	(4)	(5)	(6)
	Score		IHS(expenditure)		Score	
IHS(expenditure)	-0.087 (0.312)	0.231 (0.369)			0.023*** (0.006)	0.040*** (0.005)
Local COVID cases (Jan-Jul 2020, 10s)			0.009*** (0.003)	0.008* (0.004)		
Lagged Score		0.696*** (0.069)		0.184*** (0.015)		0.731*** (0.012)
IHS(income)	-0.009 (0.064)	-0.070 (0.319)	0.201*** (0.051)	0.860*** (0.045)	-0.032 (0.025)	0.094*** (0.014)
<i>F</i> -statistic (excl. IV)			8.91	3.10		
Student FE	✓		✓		✓	
Year FE	✓	✓	✓	✓	✓	✓
Lagged Score		✓		✓		✓
<i>N</i>	17,104	17,104	17,104	17,104	17,104	17,104

Note: The instrument is the cumulative count of confirmed COVID-19 cases through July 2020 at the si/gun/gu level (in tens of cases), from the Korea Disease Control and Prevention Agency. Columns (1)–(2) report IV/2SLS second-stage estimates, columns (3)–(4) the first stage, and columns (5)–(6) the OLS counterparts. Odd columns use student and year fixed effects (TWFE); even columns use a value-added specification with lagged test scores and year fixed effects. The excluded-instrument *F*-statistic is reported beneath the coefficients. Standard errors clustered at the school level in parentheses. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.2: IV-Style Diagnostic Using Closure-Induced Tutoring Variation

Dep. variable:	IV/2SLS		First Stage		OLS	
	(1)	(2)	(3)	(4)	(5)	(6)
	Score		IHS(expenditure)		Score	
IHS(expenditure)	-0.133 (0.123)	0.065 (0.057)			0.023*** (0.006)	0.040*** (0.005)
Closed School Days			0.041*** (0.005)	0.081*** (0.007)		
Lagged Score		0.727*** (0.011)		0.163*** (0.015)		0.731*** (0.012)
IHS(income)	-0.000 (0.034)	0.072 (0.054)	0.215*** (0.051)	0.827*** (0.042)	-0.032 (0.025)	0.094*** (0.014)
<i>F</i> -statistic			55.63	121.44		
Student FE	✓		✓		✓	
Year FE	✓	✓	✓	✓	✓	✓
Lagged Score		✓		✓		✓
<i>N</i>	17,104	17,104	17,104	17,104	17,104	17,104

Note: The instrument is closure days. Columns (1)–(2) report IV/2SLS second-stage estimates, columns (3)–(4) the first stage, and columns (5)–(6) the OLS counterparts. Odd columns use student and year fixed effects (TWFE); even columns use a value-added specification with lagged test scores and year fixed effects. Standard errors clustered at the school level in parentheses. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.3: Balance Test: Closure Days and School-Level Socioeconomic Characteristics

	(1)	(2)	(3)	(4)
	Closed School Days	Closed School Days	Closed School Days	Closed School Days
Avg IHS(income)	0.053 (0.119)			0.056 (0.118)
Avg income (10K KRW)		-0.000 (0.000)		
Avg mother's education			0.007 (0.050)	-0.004 (0.047)
<i>N</i>	542	542	539	539

Note: The unit of observation is the school. The dependent variable is closure days (in 10-day units). The sample is restricted to the COVID year (calendar year 2020). Independent variables are school-level averages of log household income and mother's education. Standard errors clustered at the city level in parentheses. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.4: Impact of Closure Days on Expenditure (Levels): by Income and Performance

	Baseline	Income			Lagged Performance		
	(1) All	(2) Low	(3) Mid	(4) High	(5) Low	(6) Mid	(7) High
Closed School Days	0.543*** (0.127)	0.577*** (0.173)	0.365* (0.206)	0.533** (0.267)	0.899*** (0.210)	0.365 (0.226)	0.273 (0.216)
p-value			0.429	0.618		0.084	0.768
N	17,104	5,886	6,338	4,880	5,702	5,702	5,700

Note: The dependent variable is monthly tutoring expenditure in 10,000 KRW. Income and lagged performance terciles are based on baseline ($g = -1$) values. All specifications include student and grade fixed effects. P-values test whether effects differ from the adjacent left column. Standard errors clustered at the school level in parentheses. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.5: Impact of School Closure Days on Parental Expenditure

	COVID Sample			All Sample					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Closed School Days	0.017*** (0.003)	-0.033 (0.028)	0.017 (0.025)	0.068*** (0.006)	0.110*** (0.010)	0.089*** (0.007)	0.040*** (0.005)	0.041*** (0.005)	0.041*** (0.006)
IHS(income)						0.891*** (0.043)		0.215*** (0.051)	0.215*** (0.051)
Above Median=1 × Closed School Days									-0.002 (0.006)
N	9,700	9,700	9,700	17,104	17,104	17,104	17,104	17,104	17,104
Grade FE		✓	✓		✓	✓	✓	✓	✓
Student FE			✓				✓	✓	✓

Note: Clustered standard errors in parentheses (school level). ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.6: Impact of School Closure Days on Test Scores

	COVID Sample			All Sample					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Closed School Days	-0.009*** (0.002)	-0.010 (0.029)	0.005 (0.017)	0.025*** (0.005)	0.048*** (0.008)	0.004 (0.005)	-0.005 (0.005)	-0.006 (0.005)	-0.004 (0.005)
IHS(tutoring expend.)						0.029*** (0.005)		0.024*** (0.006)	0.024*** (0.006)
Average Std. Score, 1-year lagged						0.746*** (0.010)			
Above Median=1 × Closed School Days									-0.005 (0.005)
<i>N</i>	9,700	9,700	11,443	17,104	17,104	14,981	17,104	17,104	17,104
Grade FE		✓	✓		✓	✓	✓	✓	✓
Student FE			✓				✓	✓	✓

Note: Clustered standard errors in parentheses (school level). ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.7: Impact of Closure Days on Tutoring Time and Quality

	Tutoring Time		Tutoring Quality	
	(1)	(2)	(3)	(4)
Closed School Days	2.009*** (0.230)	1.852*** (0.230)	-0.027*** (0.006)	-0.027*** (0.006)
IHS(income)		1.082 (2.208)		0.136** (0.060)
<i>N</i>	9,921	6,578	5,334	5,334

Note: GEPS subsample only (cohort 12). Tutoring Time is measured in hours. Tutoring Quality is defined as log(expenditure per hour). All specifications include student and grade fixed effects. Standard errors clustered at the school level in parentheses. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.8: Impact of Closure Days on Tutoring Time and Quality: by Income

	Tutoring Time				Tutoring Quality			
	(1) All	(2) Low	(3) Mid	(4) High	(5) All	(6) Low	(7) Mid	(8) High
Closed School Days	2.009*** (0.230)	2.169*** (0.331)	2.211*** (0.323)	1.812*** (0.390)	-0.027*** (0.006)	-0.031*** (0.010)	-0.032*** (0.009)	-0.018* (0.010)
p-value			0.927	0.431			0.937	0.303
N	9,921	3,159	3,627	3,135	5,334	1,590	1,968	1,776

Note: GEPS subsample only (cohort 12). Tutoring Time is measured in hours. Tutoring Quality is defined as log(expenditure per hour). Income terciles are based on 2019 household income. All specifications include student and year fixed effects. P-values test whether effects differ from the adjacent left column. Standard errors clustered at the school level in parentheses. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.9: Within-School Standard Deviation: by School Average Income

	SD(Test Score)			SD(Expenditure)		
	(1) Low	(2) Mid	(3) High	(4) Low	(5) Mid	(6) High
Closed School Days	0.021** (0.009)	0.010 (0.010)	0.004 (0.014)	-0.386 (0.330)	-0.237 (0.349)	-0.070 (0.758)
N	324	264	336	324	264	336
School FE	✓	✓	✓	✓	✓	✓
City FE	✓	✓	✓	✓	✓	✓

Note: Schools are partitioned into terciles by their average log parental income. The dependent variable is the within-school standard deviation of test scores (columns 1–3) or tutoring expenditure in 10,000 KRW (columns 4–6). Closure days are in 10-day units. All specifications include school, year, and city fixed effects. Standard errors clustered at the city level in parentheses. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.10: Impact of School Closure Days on Log Test Scores

	COVID Sample			All Sample					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Closed School Days	-0.001* (0.001)	-0.001 (0.006)	0.001 (0.002)	0.007*** (0.002)	0.012*** (0.003)	0.007*** (0.002)	-0.001 (0.001)	-0.002 (0.001)	-0.001 (0.001)
IHS(expenditure)						0.011*** (0.002)		0.006*** (0.001)	0.006*** (0.001)
Average Std. Score, 1-year lagged						0.003*** (0.000)			
Above Median=1 × Closed School Days									-0.001 (0.001)
<i>N</i>	9,700	9,700	11,443	17,164	17,164	17,164	17,164	17,164	17,164
Grade FE		✓	✓		✓	✓	✓	✓	✓
Student FE			✓				✓	✓	✓

Note: Clustered standard errors in parentheses (school level). ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.11: Impact of School Closure Days on Test Score: by Gender and Parental Education

	Baseline	Gender		Education (Mother)		Education (Father)	
	(1) All	(2) Male	(3) Female	(4) Below BA	(5) BA+	(6) Below BA	(7) BA+
Closed School Days	-0.006 (0.005)	-0.009 (0.006)	-0.003 (0.007)	-0.012** (0.006)	0.005 (0.005)	-0.014** (0.006)	0.003 (0.005)
p-value			0.561		0.029		0.032
<i>N</i>	17,104	8,874	8,214	10,282	6,336	7,926	8,642

Note: Clustered standard errors in parentheses (school level). 'BA' stands for a Bachelor's degree granted by a four-year college. P-values in columns (3), (5), and (7) test whether effects differ from their adjacent left columns. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.12: Impact of School Closure Days on Expenditure: by Gender and Parental Education

	Baseline	Gender		Education (Mother)		Education (Father)	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	All	Male	Female	Below BA	BA+	Below BA	BA+
Closed School Days	0.041*** (0.005)	0.034*** (0.007)	0.049*** (0.007)	0.042*** (0.007)	0.037*** (0.007)	0.041*** (0.008)	0.038*** (0.006)
p-value			0.135		0.654		0.762
N	17,104	8,874	8,214	10,282	6,336	7,926	8,642

Note: Clustered standard errors in parentheses (school level). P-values in columns (3), (5), and (7) test whether effects differ from their adjacent left columns. 'BA' stands for a Bachelor's degree granted by a four-year college. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.13: Impact of School Closure Days on Expenditure Stdev at School Level

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Closed School Days	0.849*** (0.145)	0.682*** (0.131)	-0.385 (0.251)	-0.269 (0.258)	-0.249 (0.246)	-0.249 (0.246)	-0.249 (0.247)
IHS(income)		23.088*** (3.535)	25.095*** (3.570)	0.598 (7.675)			0.683 (7.655)
Above Median=1 × Closed School Days				0.059 (0.240)			
N	1,206	1,206	1,206	1,206	1,206	1,206	1,206
Grade FE			✓	✓	✓	✓	✓
School FE				✓	✓	✓	✓
City FE				✓		✓	✓

Note: Clustered standard errors in parentheses (city level). ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.14: Impact of School Closure Days on Test Score Std Dev at School Level

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Closed School Days	0.000 (0.002)	0.003* (0.002)	0.019*** (0.006)	0.014* (0.007)	0.014* (0.007)	0.015** (0.007)	0.015** (0.007)
IHS(expenditure)		-0.002*** (0.000)	-0.002*** (0.000)			-0.001* (0.001)	-0.001* (0.001)
Above Median=1 × Closed School Days							0.000 (0.004)
N	1,206	1,206	1,206	1,206	1,206	1,206	1,206
Grade FE			✓	✓	✓	✓	✓
School FE				✓	✓	✓	✓
City FE					✓	✓	✓

Note: Clustered standard errors in parentheses (city level). ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.15: Quantile Regression: Tutoring Expenditure (IHS) and Test Scores

	TWFE	Quantile Regression				
	(1) Mean	(2) Q10	(3) Q25	(4) Q50	(5) Q75	(6) Q90
<i>Panel A: Dep. var. = IHS(tutoring expenditure)</i>						
Closed School Days	0.041*** (0.008)	0.255*** (0.043)	0.052*** (0.005)	0.034*** (0.003)	0.024*** (0.002)	0.023*** (0.003)
IHS(income)	0.215*** (0.054)	1.399*** (0.065)	0.725*** (0.020)	0.580*** (0.013)	0.503*** (0.010)	0.496*** (0.011)
<i>N</i>	17,104	17,104	17,104	17,104	17,104	17,104
<i>F</i> -test (equality across quantiles)	<i>F</i> = 9.14, <i>p</i> < 0.001					
<i>Panel B: Dep. var. = test score (std.)</i>						
Closed School Days	-0.006 (0.006)	-0.004 (0.006)	0.027*** (0.005)	0.044*** (0.004)	0.044*** (0.003)	0.036*** (0.003)
IHS(tutoring exp.)	0.024*** (0.006)	0.141*** (0.005)	0.175*** (0.007)	0.195*** (0.006)	0.173*** (0.009)	0.106*** (0.009)
<i>N</i>	17,104	17,104	17,104	17,104	17,104	17,104
<i>F</i> -test (equality across quantiles)	<i>F</i> = 14.28, <i>p</i> < 0.001					
Student FE	✓					
Year FE	✓	✓	✓	✓	✓	✓
Cohort FE		✓	✓	✓	✓	✓

Note: Panel A reports results for the inverse hyperbolic sine (IHS) of tutoring expenditure; Panel B reports results for standardized test scores. Column (1) reports TWFE estimates with student and year fixed effects. Columns (2)–(6) report simultaneous quantile regression (SQR) estimates with bootstrapped standard errors (200 replications), controlling for cohort and year fixed effects. The *F*-test tests the null hypothesis that the coefficient on school closure days is equal across all five quantiles. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.16: Sensitivity of Structural Estimates: TWFE $\times$ DID Penalty Weights

TWFE weight λ_{TWFE}	0	0	0	500	500	500	1500	1500	1500
Parallel-trend weight $\lambda_{\text{parallel}}$	0	250	500	0	250	500	0	250	500
<i>Structural parameters</i>									
ϕ	0.74	0.72	0.73	0.71	0.73	0.70	0.72	0.72	0.72
κ	0.43	0.41	0.41	0.43	0.41	0.44	0.42	0.41	0.42
σ_c	1.50	1.50	1.49	1.53	1.53	1.49	1.55	1.54	1.52
σ_q	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12	0.12
$\bar{\lambda}_c$	0.86	0.89	0.95	0.86	0.87	0.86	0.90	0.97	0.86
$\bar{\lambda}_q$	-0.28	-0.29	-0.29	-0.28	-0.29	-0.28	-0.28	-0.29	-0.28
<i>TWFE coefficients on closure days (data: tut= 0.041, score= -0.002, stdev= 0.270)</i>									
Log tutoring	0.006	0.029	0.021	0.028	0.041	0.042	0.033	0.050	0.046
Log test score	0.007	0.003	0.002	0.001	0.000	0.000	-0.001	-0.001	0.000
School stdev	0.300	0.297	0.391	0.296	0.285	0.290	0.279	0.273	0.277

Note: 3×3 grid of TWFE penalty weight (λ_{TWFE}) and parallel-trend penalty weight ($\lambda_{\text{parallel}}$). The parallel-trend-in-schdev weight is held fixed at $\lambda_{\text{par, sd}} = 50,000$ across all cells to isolate the TWFE $\times$ parallel-trend trade-off. Non-baseline cells are estimated from a 3% perturbation of the baseline warm start (seed varies by cell). The baseline cell ($\lambda_{\text{TWFE}}, \lambda_{\text{parallel}} = (1500, 500)$) reports the canonical estimates from the main estimation.

Table B.17: School-Level Sorting: Pre-Treatment Test Scores and SES

	(1)	(2)	(3)	(4)	(5)	(6)
	Prior test score	Prior test score	Prior test score	Prior test score	Prior test score	Prior test score
Avg IHS(income)	1.209*** (0.111)			0.737*** (0.123)		0.535*** (0.127)
Avg income (10K KRW)		0.002*** (0.000)				
Avg mother's education			0.535*** (0.048)	0.377*** (0.055)		0.377*** (0.052)
Avg IHS tutoring exp.					0.275*** (0.031)	0.156*** (0.032)
N	656	656	653	653	656	653

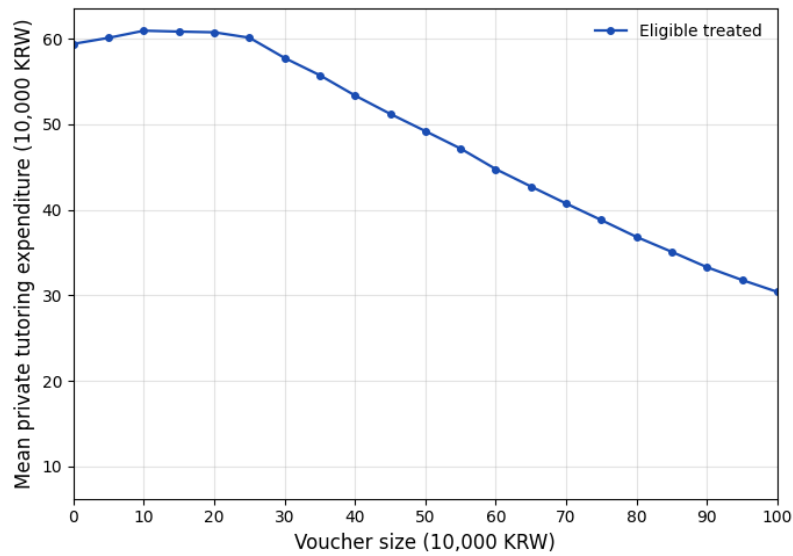
Note: The unit of observation is the school. The dependent variable in all columns is the school-level mean of pre-treatment (2019) standardized test scores. Independent variables are school-level averages of log household income, household income in 10,000 KRW, mother's years of education, and log tutoring expenditure. Robust standard errors in parentheses. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Table B.18: Quantile Regression in Levels: Tutoring Expenditure (10,000 KRW)

	TWFE	Quantile Regression				
	(1) Mean	(2) Q10	(3) Q25	(4) Q50	(5) Q75	(6) Q90
<i>Dep. var. = tutoring expenditure (10,000 KRW)</i>						
Closed School Days	0.540*** (0.121)	2.083*** (0.139)	1.691*** (0.131)	1.685*** (0.131)	1.680*** (0.129)	1.786*** (0.235)
inc	0.014*** (0.002)	0.035*** (0.001)	0.045*** (0.001)	0.055*** (0.001)	0.066*** (0.001)	0.085*** (0.002)
<i>N</i>	17,104	17,104	17,104	17,104	17,104	17,104
<i>F</i> -test (equality across quantiles)	$F = 1.50, p = 0.201$					
Student FE	✓					
Year FE	✓	✓	✓	✓	✓	✓
Cohort FE		✓	✓	✓	✓	✓

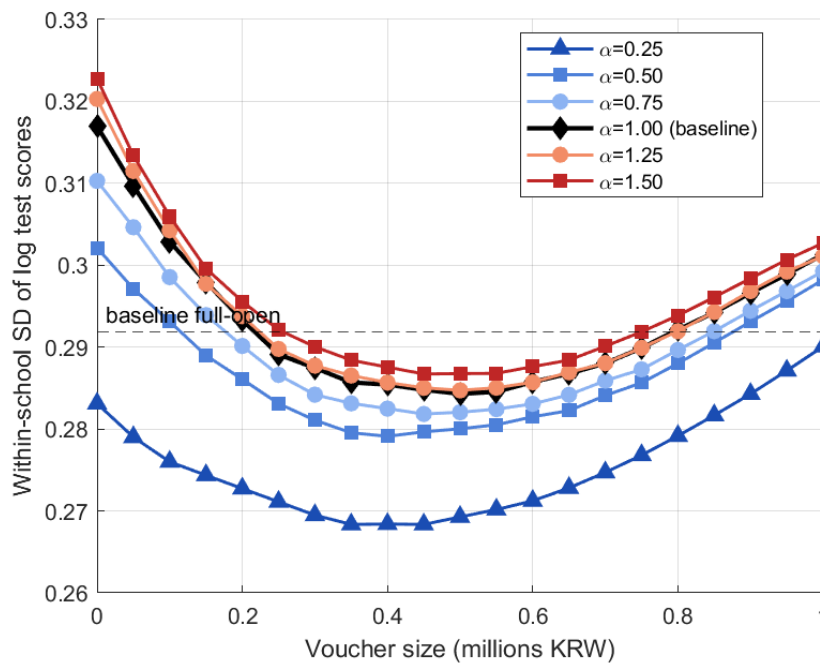
Note: Tutoring expenditure is measured in 10,000 KRW. Closure days are measured in 10-day units. Column (1) reports TWFE estimates with student and year fixed effects. Columns (2)–(6) report simultaneous quantile regression (SQR) estimates with bootstrapped standard errors (200 replications), controlling for cohort and year fixed effects. The *F*-test tests equality of the closure-days coefficient across all five quantiles. ***/**/* indicate estimate is significantly different from 0 at the 1/5/10 percent level.

Figure B.1: Voucher Crowd-Out of Private Tutoring Expenditure



Note: Mean private tutoring expenditure (in 10,000 KRW) for eligible treated households at $t = 2$, plotted against voucher size in 1,000 KRW units. Eligibility is fixed at the 60th income percentile, matching the baseline targeted voucher in Figure 8. The leftmost point ($v = 0$) corresponds to the no-voucher baseline.

Figure B.2: Sensitivity of Voucher Counterfactual to College-Prize Spread



Note: Within-school standard deviation of log test scores against voucher size at 60th-percentile eligibility. Each curve corresponds to a value of the prize-spread multiplier α applied to $\log v_j$, where $\alpha = 1.00$ is the Korean baseline. Lower α flattens the returns to college tier. Higher α steepens them. The horizontal dashed line marks the full-open inequality benchmark at $\alpha = 1.00$.

C. Structural Model Appendix

C.1 Construction of the prize term (v_j)

In this section I explain how the prize term v_j is constructed. The only difference with the Section 3.1 of Kang (2022) is that the discount factor is changed to 0.98, which reflects the change of interest rate. The prize term v_j differs by the college tiers. Conditional on the college-tier, the member colleges are assumed to share the same v_j . The tier-specific lifetime income v_j is defined as a discounted sum of the predicted income of the graduates, which is specified as

$$v_j = \sum_{t=T+1}^{T^*} \beta^{t-T} \hat{y}_{jt},$$

where $\hat{y}_{jt}$ is the income of the alumni of college tier j , and T^* is the retirement age. $\hat{y}_{jt}$ is the tier-specific annual income at time t . The predicted lifetime income $\hat{y}_{jt}$ is predicted using the regression equation,

$$\ln y_{it} = \sum_{j=1}^J (\beta_j + \delta_j \cdot age_{it}) D_{i,j}^{Tier} + Z_i \gamma + \eta_{it}, \quad (11)$$

where $D_{i,j}^{Tier}$ is the dummy variable indicating that person i graduated from a tier j college, and Z is the set of explanatory variables including squared age, birth year, and gender of person i . See Kang (2024) for the estimates of the Pooled-OLS estimates, which is used

for the prediction. The resulted vector $\mathcal{V}_{5 \times 1}$ is

$$\mathcal{V}_{5 \times 1} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{pmatrix} = \begin{pmatrix} 357554.68 \\ 295126.48 \\ 203498.74 \\ 172310.24 \\ 122007.46 \end{pmatrix}$$

C.2 Comparison: the “Max Log-score” model and the tournament model

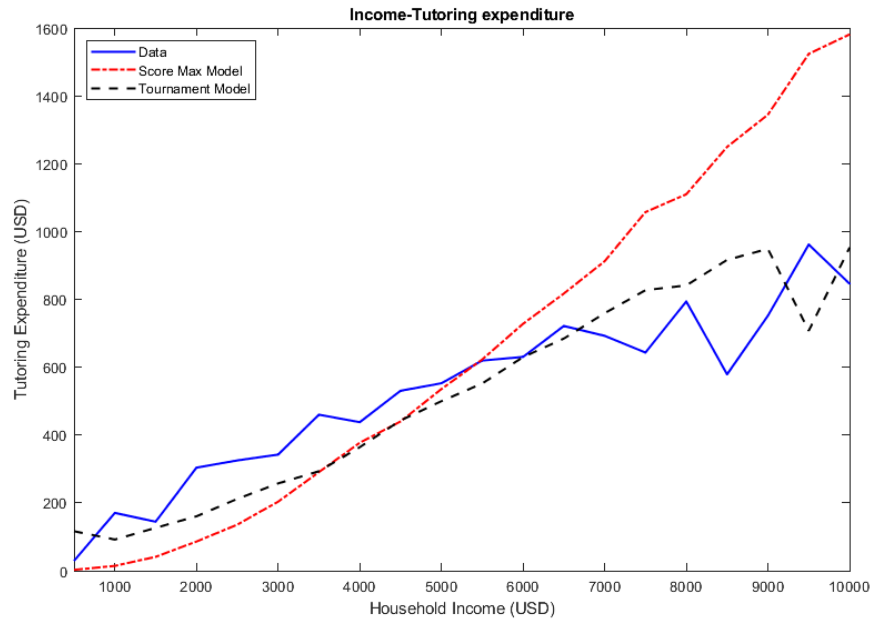


Figure C.1: Comparison with Max-Score Model

Figure C.1 presents the comparison between the “Max-score” model and the tournament model against the empirical data. The Max-score model defines the household’s terminal value as the logarithm of the final test score. While this model differs from the tournament

model only in its terminal value function (R_g), it significantly overestimates household expenditure on tutoring.

D. Likelihood Function Details

D.1 Tournament model

First Order Conditions The first order condition with respect to e_{ig} is

$$-\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} + \alpha_g \frac{\partial}{\partial e_{ig}} \sum_{j=1}^J \ln(v_j) \times Prob(\ln \tilde{Q}_{j-1} \geq \ln q_{i,g+1} \geq \ln \tilde{Q}_j \mid \Omega_{iT}, e_{ig}) = 0. \quad (12)$$

Equation 12 cannot be analytically solved in terms of e_{ig} , but the likelihood contribution is transformed from the analytical form of η_{ig}^c . Denoting the argument of the production function as B for simplicity, the first order condition g is specified as

$$\begin{aligned}
g &= -\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} + \alpha_g \left(\frac{\partial}{\partial e_{ig}} \sum_{j=1}^J \ln(v_j) \times \text{Prob}(\ln \tilde{Q}_{j-1} \geq \ln q_{i,g+1} \geq \ln \tilde{Q}_j) \right) \\
&= -\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} \\
&\quad + \alpha_g \frac{\partial}{\partial e_{ig}} \left\{ \sum_{j=1}^J \ln(v_j) \times \left(\Phi\left(\frac{\ln \tilde{Q}_{j-1} - \widehat{\ln q_{i,g+1}}}{\sigma_q}\right) - \Phi\left(\frac{\ln \tilde{Q}_{j-1} - \widehat{\ln q_{i,g+1}}}{\sigma_q}\right) \right) \right\} \\
&= -\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} \\
&\quad + \alpha_g \left[\sum_{j=1}^J \ln(v_j) \times \left(\frac{1}{\sigma_q} \phi\left(\frac{\ln \tilde{Q}_{j-1} - \widehat{\ln q_{i,g+1}}}{\sigma_q}\right) - \frac{1}{\sigma_q} \phi\left(\frac{\ln \tilde{Q}_j - \widehat{\ln q_{i,g+1}}}{\sigma_q}\right) \right) \left(-\frac{\nu_g \delta_{eg} (1 + e_{ig})^{\phi_g - 1}}{B} \right) \right] \\
&= -\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} \\
&\quad + \alpha_g \left[\ln(v_1) \frac{1}{\sigma_q} \phi\left(\frac{\ln \tilde{Q}_0 - \widehat{\ln q_{ig}}}{\sigma_q}\right) - \frac{1}{\sigma_q} \sum_{j=1}^{J-1} \left(\ln(v_j) - \ln(v_{j+1}) \right) \phi\left(\frac{\ln \tilde{Q}_j - \widehat{\ln q_{ig+1}}}{\sigma_q}\right) - \frac{1}{\sigma_q} \ln(v_J) \right. \\
&\quad \left. \phi\left(\frac{\ln \tilde{Q}_J - \widehat{\ln q_{ig+1}}}{\sigma_q}\right) \right] \times \left(-\frac{\delta_3 (1 + e_{ig})^{\phi_g - 1}}{B} \right),
\end{aligned}$$

where $B = \delta_q q_{ig}^\phi + \delta_s \text{Schdays}_{ig}^\phi + \delta_e (1 + e_{ig})^\phi$

$$\begin{aligned}
g &= -\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} + \alpha_g \left(\frac{\partial}{\partial e_{ig}} \sum_{j=1}^J \ln(v_j) \times \text{Prob}(\ln \tilde{Q}_{j-1} \geq \ln q_{i,g+1} \geq \ln \tilde{Q}_j) \right) \\
&= -\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} + \alpha_g \frac{\partial}{\partial e_{ig}} \left[\sum_{j=1}^J \ln(v_j) \times \left\{ \Phi\left(\sigma_q^{-1} \ln \frac{\tilde{Q}_{j-1}}{q_{i,g+1}}\right) - \Phi\left(\sigma_q^{-1} \ln \frac{\tilde{q}_{i,g+1}}{\tilde{Q}_j}\right) \right\} \right] \\
&= -\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} \\
&\quad + \alpha_g \left[\sum_{j=1}^J \ln(v_j) \times \sigma_q^{-1} \left\{ \phi\left(\sigma_q^{-1} \ln \frac{\tilde{Q}_{j-1}}{q_{i,g+1}}\right) - \phi\left(\sigma_q^{-1} \ln \frac{\tilde{q}_{i,g+1}}{\tilde{Q}_j}\right) \right\} \left(-\frac{\nu_g \delta_{eg} (1 + e_{ig})^{\phi_g - 1}}{B} \right) \right] \\
&= -\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} + \alpha_g \left[\ln(v_1) \frac{1}{\sigma_q} \phi\left(\frac{\ln \tilde{Q}_0 - \widehat{\ln q_{ig}}}{\sigma_q}\right) - \frac{1}{\sigma_q} \sum_{j=1}^{J-1} \left(\ln(v_j) - \ln(v_{j+1}) \right) \phi\left(\frac{\ln \tilde{Q}_j - \widehat{\ln q_{ig+1}}}{\sigma_q}\right) - \frac{1}{\sigma_q} \ln(v_J) \right. \\
&\quad \left. \phi\left(\frac{\ln \tilde{Q}_J - \widehat{\ln q_{ig+1}}}{\sigma_q}\right) \right] \times \left(-\frac{\delta_3 (1 + e_{ig})^{\phi_g - 1}}{B} \right),
\end{aligned}$$

Since $\ln \tilde{Q}_0 = \infty$ and $\ln(\tilde{Q}_j) = -\infty$, the first order condition H is specified as follows.

$$\begin{aligned} H &= -\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} + \alpha_g \left[-\sum_{j=1}^{J-1} \left(\ln(v_j) - \ln(v_{j+1}) \right) \frac{1}{\sigma_q} \phi\left(\frac{\ln \tilde{Q}_j - \ln \widehat{q_{i,g+1}}}{\sigma_q}\right) \right] \left(-\frac{\nu_g \delta_g (1 + e_{ig})^{\phi_g - 1}}{B} \right) \\ &= -\frac{\eta_{ig}^c}{w_{ig} - e_{ig}} + \alpha_g \underbrace{\left[\sum_{j=1}^{J-1} \left(\ln(v_j) - \ln(v_{j+1}) \right) \frac{1}{\sigma_q} \phi\left(\frac{\ln \tilde{Q}_j - \ln \widehat{q_{i,g+1}}}{\sigma_q}\right) \right]}_{\varphi} \left(\frac{\delta_g (1 + e_{ig})^{\phi_g - 1}}{B} \right) = 0 \end{aligned}$$

The likelihood contribution of e_{ig} is transformed from the specified log-shock $\tilde{\zeta}_{ig}^c \equiv \ln \tilde{\eta}_{ig}^c$ that makes $H = 0$. In particular,

$$\tilde{\zeta}_{ig}^c = \ln \alpha_g + \ln \delta_{eg} + \log \varphi + (\phi_g - 1) \ln(1 + e_{ig}) + \ln(w_{ig} - e_{ig}) - \ln B.$$

On the other hand, the log test-score shock at its particular point is

$$\tilde{\zeta}_{ig}^q = \ln q_{i,g+1} - \left(\delta_{0g} + \frac{1}{\phi_g} \ln(\delta_{qg} q_{ig}^{\phi_g} + \delta_{sg} schdays_{ig}^{\phi_g} + \delta_{eg} (1 + e_{ig})^{\phi_g}) \right).$$

D.2 Likelihood Evaluation details

Individual likelihood contribution The individual likelihood contribution is composed of the contributions regarding private tutoring expenditure e_{ig} and test score $q_{i,g+1}$.

For the test scores, the likelihood contribution is a probability density function (PDF) for all individuals. For private tutoring expenditure, the likelihood contribution differs depending upon whether the household participates in private tutoring activities. For the participants, likelihood contribution is a PDF since tutoring expenditure is a positive continuous variable. On the other hand, the non-participants' likelihood contribution is a cumulative distribution function, as it needs to integrate all possible shocks that make individual not participate. Denoting Θ_g^c as the set of parameters, each household

i 's likelihood contribution is

$$\mathcal{L}_i(\Theta_g^c | \{q_{ig}\}_g^{g+1}, \{w_{ig}\}_g^{g+1}) = \prod_g^{g+1} \int_{\eta_{ig}^q} \mathcal{L}_{ig}(\theta | q_{ig}, w_{ig}) f(\eta_{ig}^q) d\eta_{ig}^q, \quad (13)$$

and

$$\begin{aligned} \mathcal{L}_{ig}(\Theta_g^c | q_{ig}, w_{ig}) &= \left[f_{e_{ig}}(e_{ig}) \cdot f_{q_{i,g+1}}(q_{i,g+1} | e_{ig}) \right]^{d_{ig}^e} \\ &\quad \times \left[\Pr(e_{ig} = 0) \cdot f_{q_{i,g+1}}(q_{i,g+1} | e_{ig}) \right]^{(1-d_{ig}^e)}; \end{aligned}$$

where d_{ig}^e is an indicator function of tutoring participation decision (i.e. $e_{ig} > 0$ if $d_{ig} = 1$).

The unrestricted part of the likelihood function is

$$\mathcal{L}(\Theta_g^c) = \prod_{i=1}^N \mathcal{L}_i(\theta_g^c | \{q_{ig}\}_g^{g+1}, \{w_{ig}\}_g^{g+1}).$$

Jacobian Transformation We denote the log-shocks by $\zeta_{ig}^z = \ln \eta_{ig}^z$ for $z = c, q$, and their particular points at the maximizing tutoring decision by $\tilde{\zeta}_{ig}^z = \ln \tilde{\eta}_{ig}^z$. For the participant of the tutoring expenditure, the Jacobian transformation from the log-shocks (ζ^c, ζ^q) to the observables $(e_{ig}, q_{i,g+1})$ is

$$\begin{aligned} |J_1| &= \det \left| \frac{\partial(\tilde{\zeta}_{ig}^c, \tilde{\zeta}_{ig}^q)}{\partial(e_{ig}, q_{i,g+1})} \right| \\ &= \frac{\partial \tilde{\zeta}_{ig}^c}{\partial e_{ig}} \frac{\partial \tilde{\zeta}_{ig}^q}{\partial q_{i,g+1}} - \frac{\partial \tilde{\zeta}_{ig}^q}{\partial e_{ig}} \frac{\partial \tilde{\zeta}_{ig}^c}{\partial q_{i,g+1}} \\ &= \frac{\partial \tilde{\zeta}_{ig}^c}{\partial e_{ig}} \frac{\partial \tilde{\zeta}_{ig}^q}{\partial q_{i,g+1}} \end{aligned}$$

From $\frac{\partial}{\partial x} \frac{1}{\sigma} \phi\left(\frac{x}{\sigma}\right) = -\frac{x}{\sigma^2} \left(\frac{1}{\sigma} \phi\left(\frac{x}{\sigma}\right)\right)$, we have

$$\frac{\partial \zeta_{ig}^c}{\partial e_{ig}} = \frac{1}{\varphi} \frac{\partial \varphi}{\partial e_{ig}} + \frac{\phi - 1}{(1 + e_{ig})} - \frac{1}{w_{ig} - e_{ig}} - \frac{1}{\delta_q q_{ig}^\phi + \delta_s Schdays_{ig}^\phi + \delta_e (1 + e_{ig})^\phi} \delta_e \phi (1 + e_{ig})^{(\phi-1)},$$

where $\varphi = \sum_{j=1}^{J-1} \left(\ln(v_j) - \ln(v_{j+1}) \right) \frac{1}{\sigma_q} \phi \left(\frac{\ln \tilde{Q}_j - \ln \widehat{q_{i,g+1}(e_{ig})}}{\sigma_q} \right)$ and

$$\begin{aligned} \frac{\partial \varphi}{\partial e_{ig}} &= \sum_{j=1}^{J-1} \left(\ln(v_j) - \ln(v_{j+1}) \right) \left\{ -\frac{\ln \tilde{Q}_j - \ln \widehat{q_{i,g+1}(e_{ig})}}{\sigma_q^2} \right\} \frac{1}{\sigma_q} \phi \left(\frac{\ln \tilde{Q}_j - \ln \widehat{q_{i,g+1}(e_{ig})}}{\sigma_q} \right) \left(\frac{-\partial \ln \widehat{q_{i,g+1}(e_{ig})}}{\partial e_{ig}} \right) \\ &= \sum_{j=1}^{J-1} \left(\ln(v_j) - \ln(v_{j+1}) \right) \left\{ \frac{\ln \tilde{Q}_j - \ln \widehat{q_{i,g+1}(e_{ig})}}{\sigma_q^2} \right\} \frac{1}{\sigma_q} \phi \left(\frac{\ln \tilde{Q}_j - \ln \widehat{q_{i,g+1}(e_{ig})}}{\sigma_q} \right) \left(\frac{\partial \ln \widehat{q_{i,g+1}(e_{ig})}}{\partial e_{ig}} \right) \\ &= \sum_{j=1}^{J-1} \left\{ \left(\ln(v_j) - \ln(v_{j+1}) \right) \left\{ \frac{\ln \tilde{Q}_j - \ln \widehat{q_{i,g+1}(e_{ig})}}{\sigma_q^2} \right\} \frac{1}{\sigma_q} \phi \left(\frac{\ln \tilde{Q}_j - \ln \widehat{q_{i,g+1}(e_{ig})}}{\sigma_q} \right) \right\} \left(\frac{\delta_e \phi (1 + e_{ig})^{(\phi-1)}}{B} \right). \end{aligned}$$

For the non-participants, the Jacobian transformation term is

$$\begin{aligned} |J_2| &= \det \left| \frac{\partial \tilde{\zeta}_{ig}^q}{\partial q_{i,g+1}} \right| \\ &= \frac{\partial \tilde{\zeta}_{ig}^q}{\partial q_{i,g+1}} \end{aligned}$$

The likelihood contribution uses the equilibrium conditions of the theoretical framework. I denote $\tilde{\eta}_{ig}^c$ and $\tilde{\eta}_{ig}^q$ as the particular points of the level shocks where the utility of the household is maximized, and $\tilde{\zeta}_{ig}^z = \ln \tilde{\eta}_{ig}^z$ as the corresponding particular points of the log-shocks for $z = c, q$. The log-shocks (ζ^c, ζ^q) are assumed to be jointly normal, and the likelihood function is evaluated on the log scale. In particular,

$$\begin{aligned} \mathcal{L}_{ig}(\Theta_g^c | q_{ig}, w_{ig}) &= \left[f_{\zeta_{ig}^c}(\tilde{\zeta}_{ig}^c) \cdot f_{\zeta_{ig}^q}(\tilde{\zeta}_{ig}^q | \zeta_{ig}^c) |J_i^1| \right]^{d_{ig}} \\ &\times \left[\int_{\tilde{\zeta}_{ig}^c}^{\infty} f_{\zeta_{ig}^c}(\zeta_{ig}^c) \cdot f_{\zeta_{ig}^q}(\tilde{\zeta}_{ig}^q | \zeta_{ig}^c) d\zeta_{ig}^c |J_i^2| \right]^{(1-d_{ig}^e)} \end{aligned}$$

where $|J_i^j|_{j=1,2}$ is the corresponding Jacobian transformation term. For the non-participants of the tutoring activities, they do not spend on tutoring expenditure if their log-shock is above the threshold $\tilde{\zeta}_{ig}^c$, which is the minimum value of the log consumption shock that makes the household stop spending on tutoring expenditure. Note that θ_g^c depends on the student's grade g and the cohort the student belongs to.

D.3 Structural Estimates

Table D.1: Grade-Specific Structural Parameters

		8th Grade		11th Grade		12th Grade	
		Control	Treated	Control	Treated	Control	Treated
Tutoring Effects (δ_e)	g	0.253 (0.0009)	0.383 (0.0009)	0.386 (0.0009)	0.447 (0.0008)	0.371 (0.0008)	0.437 (0.0007)
	$g + 1$	0.310 (0.0004)	0.332 (0.0004)	0.313 (0.0006)	0.402 (0.0005)	0.319 (0.0006)	0.292 (0.0005)
School Days Effects (δ_s)	g	0.106	0.161	0.162	0.187	0.156	0.183
	$g + 1$	0.130	0.139	0.131	0.168	0.134	0.122
Altruism (α)	g	5.368 (0.168)	4.352 (0.197)	1.977 (0.081)	5.227 (0.203)	1.727 (0.052)	5.307 (0.153)
	$g + 1$	0.087 (0.003)	0.010 (0.000)	1.102 (0.045)	9.786 (0.274)	0.411 (0.017)	3.353 (0.107)
Previous Test Scores (δ_0)	g	0.707 (0.0009)	0.799 (0.0010)	0.677 (0.0009)	0.693 (0.0010)	0.659 (0.0007)	0.601 (0.0007)
	$g + 1$	0.233 (0.0005)	0.186 (0.0006)	0.291 (0.0006)	0.533 (0.0006)	0.256 (0.0005)	0.360 (0.0006)

Note: Standard errors in parentheses computed using delta method. The effects of school days are implied estimates using estimated κ , thus indicated by “-”.

Table D.2: Year-Specific Fit: Expenditure and Log Test Scores (Data vs. Model)

	Data		Model	
	2019	2020	2019	2020
Panel A: Tutoring Expenditure				
Treated 12th:	70.37	75.63	71.97	74.87
Control 12th:	53.63	55.10	53.93	54.38
Treated 11th:	62.41	64.31	63.35	63.20
Control 11th:	55.52	58.55	55.74	57.39
Treated 8th:	43.16	49.41	41.25	53.90
Control 8th:	35.46	37.94	36.93	36.12
Panel B: Log Test Scores				
Treated 12th:	4.68	4.66	4.97	4.94
Control 12th:	4.51	4.51	4.95	4.80
Treated 11th:	4.63	4.63	5.01	4.96
Control 11th:	4.60	4.61	4.99	4.88
Treated 8th:	4.63	4.63	5.04	4.68
Control 8th:	4.62	4.62	5.03	4.77

Note: Unit of tutoring expenditure is 10,000KRW $\approx$ 8 USD. Each cohort is named after their grade in $g = 0$. For example, 12th grade cohort refers to students who were in 12th grade at $g = 0$)